

LAB 3 : TokTickIT Authenticated IT Ticketing System

Final submission report - authenticated users, IT Staff ticketing, Administrator safety, and release evidence

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Answer Part 1: Git Use with Engineering Workflow (10 Points)

Lab 3 was developed on issue-specific branches, reviewed through pull requests, integrated into lab3-staging, and released to main. The final release is PR #46 at commit 92fac36.

Evidence	Link
GitHub repository	HolyThiccDaddy/toktickit
GitHub project board	TokTickIT Individual Sprints
Issue #36 - authentication and migration	Issue #36
Issue #37 - requester authorization regression	Issue #37
Issue #38 - staff queue and ticket operations	Issue #38
Issue #39 - Administrator and release evidence	Issue #39
Final release PR #46	PR #46 - merged into main

Final release status: PR #46 was peer-approved and merged into main as 92fac36. The documentation follow-up in this report is submitted separately as PR #47.

Answer Part 1: Git Use with Engineering Workflow (continued)

Peer-review and release history

Stage	Pull request and result
Contract	PR #40 approved and merged as 40ca398.
Authentication and migration	PR #41 merged as b49c8b2 after credential and path-gate fixes.
Requester regression	PR #42 merged as 0ee0002.
Staff queue	PR #43 merged as c1cd708.
Administrator implementation	PR #44 approved after requested changes and merged as bd5872a.
Documentation follow-up	PR #45 approved and merged as e82ac21.
Final integration	PR #46 approved and merged into main as 92fac36.

The reviewer record identifies requested changes, responses, approval reviews, and merge commits. The final board and branch state are recorded in the source documents linked in Parts 2 and 3.

Answer Part 2: Spec DD (5 Points)

Approved engineering contract and traceability

Document	Purpose and evidence
Specification	docs/lab-03/specification.md - requirements, roles, business rules, data model, migration, and Definition of Done.
API contract	docs/lab-03/api-spec.md - named request/response schemas, status codes, authorization, and compatibility.
UI specification	docs/lab-03/ui-spec.md - screens, states, Zen Green tokens, accessibility, and responsive rules.
Test plan	docs/lab-03/tests.md - acceptance-criterion traceability and real execution evidence.

Contract-before-implementation evidence

PR #40 records the approved contract before implementation. It covers authentication, first-login password change, role and ownership authorization, Lab 2 migration, staff queue/detail, comments and notes, Administrator safeguards, responsive UI, and acceptance-criterion tests.

The contract is extended by the final release record rather than replaced: the merged main commit is 92fac36, and the final PDF is generated from the source tree at that release.

Key contract decisions

- Three authenticated roles: REQUESTER, IT_STAFF, and ADMINISTRATOR, with server-side authorization.
- The first-login password gate applies globally to protected APIs until the new password is saved.
- Lab 2 requester ownership and identifiers are preserved through migration and idempotent seed.
- IT Priority, complete status transitions, public comments, and staff-only Internal Notes use named schemas and tests.
- Administrator account safety prevents duplicate emails, self-deactivation, and removal of the last active Administrator.

Answer Part 3: Test DD and Traceability (10 Points)

Verification results from the merged Lab 3 release

Check	Executed result
Server regression	15 test files / 92 tests passed twice against the dedicated _test database.
Client regression	15 test files / 50 tests passed twice; the Create Ticket loading race is stabilized.
Server production build	TypeScript production build passed.
Client production build	TypeScript and Vite production build passed.
Playwright E2E	13 tests passed across desktop, tablet, and mobile projects.
Migration and seed	Seven migration/regression tests passed, including preserved Lab 2 rows, collision-safe IDs, pending credentials, and deterministic reseeding.
Repository hygiene	git diff --check passed for source documentation; local environment files and credentials remain ignored.

Acceptance-criterion traceability

Criteria	Evidence mapped in tests.md
AC-01 to AC-04	Login, first-login password gate, logout/expiry, CSRF, and role-aware session APIs and UI tests.
AC-05 to AC-06	Requester ownership, comments, resolution indication, attachment compatibility, and regression E2E.
AC-07 to AC-09	Shared queue, staff detail, assignment, IT Priority, status matrix, public comments, and Internal Notes.
AC-10 to AC-13	Administrator listing/filtering, create/edit/activation/reset, safety protections, migration, and forged-identity rejection.
AC-14 to AC-15	Responsive screenshots, visual checks, complete regression suites, production builds, and release integration.

Detailed test-file paths and command results are maintained in [docs/lab-03/tests.md](https://github.com/department-of-homeland-security/doc-examples/blob/master/docs/lab-03/tests.md).

Answer Part 4: AI Use with Reflection (5 Points)

LLM used: OpenAI Codex (GPT-5). The agent output was treated as a draft and accepted only after comparison with Lab_3_sheet.pdf, the approved contract, executable tests, real terminal evidence, and peer-review feedback.

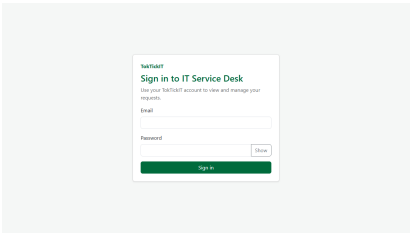
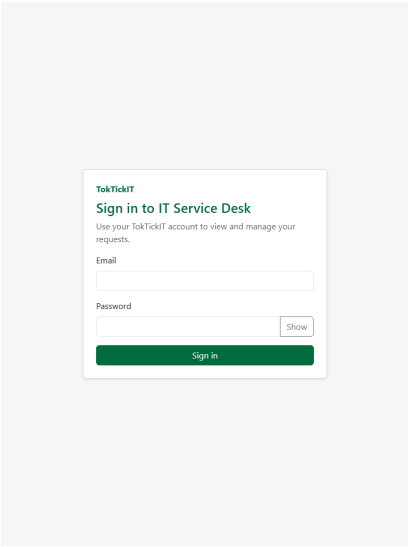
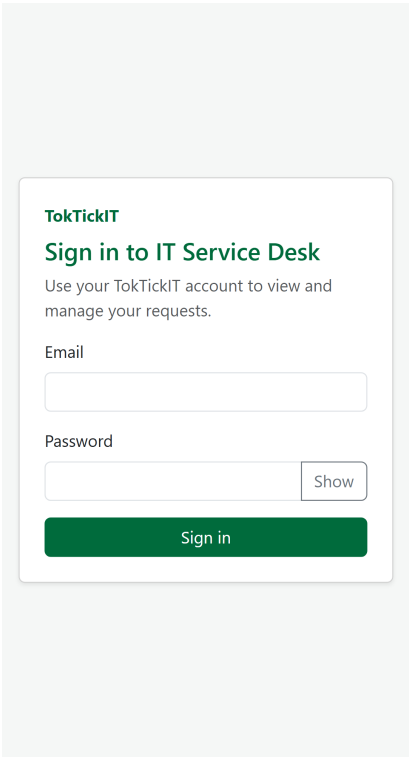
#	Selected prompt and accepted result
1	Convert the Lab 3 handout into a traceable product increment, issue plan, and Definition of Done.
2	Draft the engineering contract for authentication, role authorization, migration, queue/detail, comments/notes, and Administrator safety.
3	Implement the server foundation with sessions, password gate, CSRF, migration compatibility, and regression tests.
4	Implement requester authorization and communication while preserving Lab 2 ownership and attachment behavior.
5	Design Staff Queue, Ticket Detail, User Management states, responsive layouts, and safe failure feedback.
6	Map every acceptance criterion to unit, API, UI, security, migration, responsive, and E2E checks.
7	Recheck peer-review comments for contract, migration, authorization, concurrency, filtering, and E2E evidence gaps.
8	Audit release evidence for source/test-count consistency and final-main readiness.

My Reflection

Codex was most useful for turning the handout into a traceable contract, finding mismatches between requirements and tests, and generating focused regression cases. Human decisions and peer review remained authoritative: authorization, migration safety, password handling, and status transitions were accepted only after executable tests and real evidence passed. Suggestions that weakened server-side checks, exposed credentials, relied on hidden UI controls, or added excluded features were changed or rejected. The final release was rechecked from merged main before this report was generated.

Answer Part 5: Working Login and Password Change UI (5 Points)

The authenticated session boundary replaces the Lab 2 Development Requester selector. The UI covers valid and invalid login, inactive-account handling, first-login password change, role-aware navigation, logout, and direct-access protection after logout.

Authentication and first-login password-change evidence		
		
Desktop	Tablet	Mobile

Real Playwright captures from the merged Lab 3 release. Desktop, tablet, and mobile views show the same Zen Green form hierarchy without clipping or horizontal overflow.

Supporting tests: server/tests/lab-03/auth.api.test.ts, authorization.api.test.ts; client/tests/lab-03/Login.test.tsx and ChangePassword.test.tsx; e2e/lab-03/authentication.spec.ts.

Answer Part 6: Working IT Staff Ticket Queue UI (5 Points)

IT Staff and Administrators share a searchable queue with status, IT Priority, assignee, category, sorting, pagination, realistic empty/no-results states, retryable failure feedback, and a responsive card layout.

IT Staff Ticket Queue responsive evidence

TokTicketIT

Ticket Queue

Alex Staff
IT_STAFF

Sign out

Ticket Queue

Shared queue for IT Staff.

IT_STAFF

Search queue

Ticket, summary, requester, or ca

Status

All statuses

IT Priority

All priorities

Assignee ID

Any assignee

Category

All categories

Sort by

Updated date

Sort direction

Descending

Results per page

20

Clear

Ticket No.	Updated	Status	Summary	Requester	IT Priority	Assignee	Actions
TKT-2026-000001	9/17/2026, 4:48:58 PM	NEW	Issue 39 visual desktop	Sarah Johnson	HIGH	Unassigned	Open

1 result · Page 1 of 1

TokTicketIT

Ticket Queue

Alex Staff
IT_STAFF

Sign out

Ticket Queue

Shared queue for IT Staff.

IT_STAFF

Search queue

Ticket, summary, requester, or category

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All statuses

IT Priority

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Assignee ID

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Clear

Ticket No.	Updated	Status	Summary	Requester	IT Priority	Assignee	Actions
TKT-2026-000001	9/17/2026, 4:48:58 PM	NEW	Issue 39 visual tablet	Sarah Johnson	HIGH	Unassigned	Open

1 result · Page 1 of 1

TokTicketIT

Ticket Queue

Alex Staff
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Sign out

Ticket Queue

Shared queue for IT Staff.

IT_STAFF

Search queue

Ticket, summary, requester, or category

Status

All statuses

IT Priority

All priorities

Assignee ID

Any assignee

Category

All categories

Sort by

Updated date

Sort direction

Descending

Results per page

20

Clear

TKT-2026-000002

NEW

Issue 39 visual mobile

Requester: Sarah Johnson

HIGH Unassigned

Updated 9/17/2026, 4:49:14 PM

Open ticket

TKT-2026-000001

NEW

Issue 39 visual tablet

Requester: Sarah Johnson

HIGH Unassigned

Updated 9/17/2026, 4:48:58 PM

Open ticket

2 results · Page 1 of 1

Previous

1

Next

Desktop

Tablet

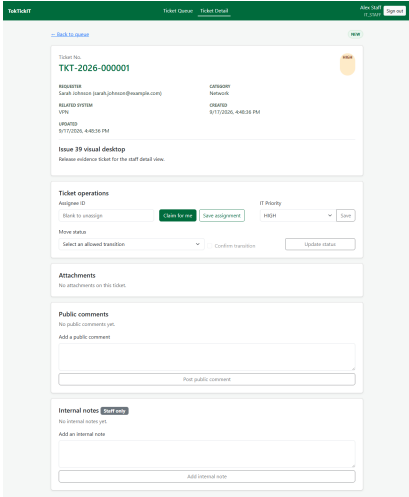
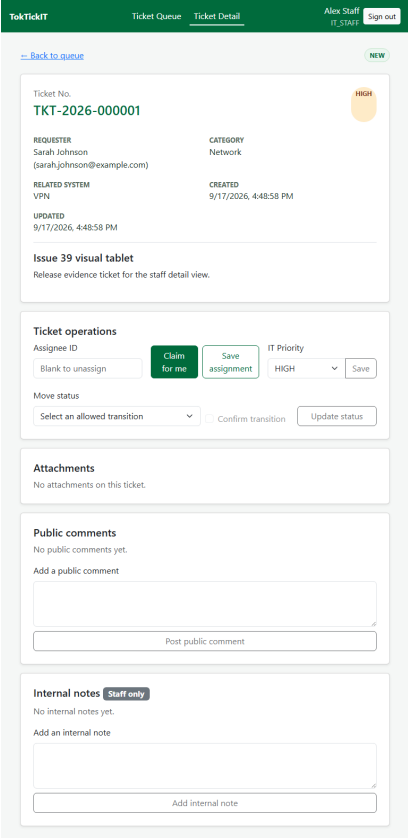
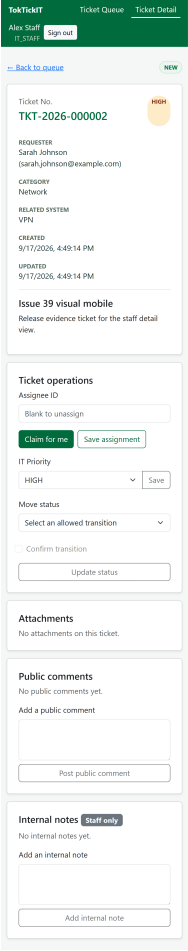
Mobile

The mobile card layout includes visible sorting controls, while desktop and tablet views show the table and filter hierarchy. The Open Detail action is backed by the staff detail route and regression coverage.

Supporting tests: server/tests/lab-03/staff-queue.api.test.ts; client/tests/lab-03/StaffTicketQueue.test.tsx; e2e/lab-03/staff-ticket-flow.spec.ts.

Answer Part 7: Working IT Staff Ticket Detail UI (10 Points)

Staff Ticket Detail supports claim/reassign, IT Priority, permitted status transitions and confirmations, attachment continuity, Public Comments, Internal Notes, Requester resolution indication, role restrictions, validation, and safe failure behavior.

IT Staff Ticket Detail responsive evidence		
		
Desktop	Tablet	Mobile

The three real-run captures show the ticket summary, operations, attachments, Public Comments, and staff-only Internal Notes at desktop, tablet, and mobile widths.

Supporting tests: server/tests/lab-03/staff-queue.api.test.ts and comments-notes.api.test.ts; client/tests/lab-03/StaffTicketDetail.test.tsx; e2e/lab-03/staff-ticket-flow.spec.ts.

The Administrator-only screen lists users, supports name/email search and role filtering, creates and edits one-role accounts, changes activation state, sets an initial password, and enforces duplicate-email and last-Administrator protections.

The image displays three versions of a user management interface for a system called TokTicket, designed for Desktop, Tablet, and Mobile screens.

Desktop View: The interface features a top navigation bar with 'TokTicket', 'User Management', 'Ticket Queue', and user roles 'ADMIN' and 'Sign out'. The main heading is 'User Management' with a subtitle 'Manage roles and account activation without exposing credentials.' and an 'ADMIN' badge. A search bar allows filtering by 'Name or email' and 'All roles'. A table lists users with columns for Display name, Email, Role, State, and Actions. The table includes 12 rows of user data, with roles like IT_STAFF, REQUESTER, and ADMIN. The Actions column contains buttons for Edit, Deactivate, and Reset password.

Tablet View: The layout is more compact, with the search bar and role filter integrated into a single row. The table columns remain the same, but the font size and button spacing are adjusted for a medium screen. The 'ADMIN' badge is still present.

Mobile View: The interface is highly condensed. The search bar and role filter are stacked vertically. The table is simplified, with the Role and State columns being less prominent. The Actions column buttons are smaller and more closely spaced. The 'ADMIN' badge is located at the bottom right of the table.

Supporting tests: `server/tests/lab-03/users-admin.api.test.ts` and `authorization.api.test.ts`; `client/tests/lab-03/UserManagement.test.tsx`; `e2e/lab-03/user-administration.spec.ts`.

Answer Part 9: Zen Green UI and Responsive Evidence (5 Points)

The released screens reuse the Zen Green tokens and shared controls established in Lab 2. Visual tests and Playwright projects cover desktop, tablet, and mobile layouts for authentication, User Management, Staff Queue, and Staff Ticket Detail.

View	Desktop	Tablet	Mobile
Authentication	Login hierarchy, password visibility, safe errors	Scaled form with stable spacing	Single-column form without overflow
User Management	Table/list actions and filters	Readable list with preserved actions	Responsive cards and filter controls
Staff Queue	Table, sortable columns, pagination	Filter layout remains usable	Sortable cards with visible controls
Staff Ticket Detail	Operations and communications panels	Panels stack without clipping	Readable stacked controls and notes

Completed visual checklist

- Zen Green colors, typography, field styling, and button hierarchy are consistent across role shells.
- Loading, empty/no-results, validation, failure/retry, and submitting states are represented.
- Focus order, labels, safe error messages, readable badges, and editable/read-only boundaries are checked.
- No clipping, overlap, horizontal overflow, or hidden mobile queue sorting controls remain.

Submission status: contract, tests, AI-use record, reviewer history, responsive evidence, and final release are complete. Final Release PR #46 is merged into main at 92fac36; this report is the concise PDF required by the Lab 3 handout.

Lab 3 exclusions: external identity providers, password-recovery email, multi-role accounts, staff assignment history, and administrative deletion remain outside the approved scope. Deactivation is used instead of deletion so historical ownership and authored communication remain queryable.

LAB 3 : Final Evidence Addendum

TokTickIT Authenticated IT Ticketing System — Lab 2 template continuation

The first ten pages of this report use the original Lab 2 submission layout: centered title, student-information table, underlined Answer Parts, gray evidence tables, yellow status callouts, and numbered footer. This addendum keeps that same visual language while adding the final contract crosswalk and local verification captures.

Release provenance

Implementation release: **main @ 92fac36**, merged by PR #46. Documentation/evidence follow-up: **e7998d3**, PR #47. The retained terminal frames show the final checkout path and keep sensitive database values hidden.

The report uses vector ReportLab text and tables throughout the template pages and addendum. Screenshot evidence is embedded at its source PNG resolution; it is not first rasterized into a low-resolution page image.

Contract completeness crosswalk

Required contract element	Evidence in approved source	Release check
Sprint goal, stakeholder request, and scope/exclusions	specification.md §§1–3	Scope is explicit before implementation
Numbered functional requirements and business/security rules	specification.md §§4–5 (FR-01–FR-16, BR-01–BR-14)	Authentication, ownership, CSRF, status, migration, and safe-error rules are testable
UI specification summary	specification.md §6 plus ui-spec.md	Shell, screens, states, roles, breakpoints, accessibility, and no-overflow rules are named
Authorization matrix	specification.md §7	Requester, IT Staff, and Administrator capabilities are explicit; Administrator Ticket access is an intentional grant
Data/migration decisions	specification.md §8/§8.1	Existing Lab 2 identifiers/relationships and deterministic credential backfill are preserved
API contract	specification.md §9 plus api-spec.md	Envelopes, schemas, status codes, CSRF/session boundary, authorization, and safe errors are named
Acceptance criteria and Definition of Done	specification.md §§10–11	AC-01–AC-15 map to executable checks in tests.md
Review assumptions	specification.md §12	Session, crypto, role, migration-password, and report assumptions are recorded

This crosswalk closes the common Spec DD gap where API and UI documents are linked separately but the master specification does not show how the complete contract is assembled.

Test and traceability proof

Gate	Verified result and evidence
Server regression	16 test files / 92 tests passed after the explicit Staff Ticket Detail suite split; Lab 3 API, security, migration, queue/detail, comments/notes, and admin paths are named in tests.md
Client regression	15 test files / 50 tests passed from .capture-main in the retained final terminal capture; Login, ChangePassword, requester detail, queue, staff detail, admin, and responsive suites
Builds	server npm run build and client npm run build passed; terminal evidence items 26–27
Playwright	13 tests passed across desktop/tablet/mobile; authentication, requester regression, staff flow, admin, and responsive visual suites
Migration/seed	Guarded test database, migrate deploy, and two deterministic seed runs; terminal evidence item 30
Traceability	tests.md maps AC-01–AC-15 to exact API/UI/security/migration/E2E paths and names the staff-detail assertions in the shared staff queue API suite

Staff queue and Staff Ticket Detail now have explicit API test files. They share a fixture helper so the suite remains deterministic while the repository structure matches the Lab 3 handout.

Part 5–9 full-credit evidence map

Part	Visual evidence	Executable / contract evidence
5 Login/password	authentication desktop/tablet/mobile captures	auth + authorization API, Login + ChangePassword UI, authentication E2E; valid, invalid, inactive, first-login, logout, and direct API gate
6 Staff queue	staff-queue desktop/tablet/mobile captures	queue API/UI/E2E; query, filters, sort, page controls, empty/no-results, retry, and mobile card sorting
7 Staff detail	staff-ticket-detail desktop/tablet/mobile captures	queue/detail API assertions, StaffTicketDetail UI, comments/notes visibility, assignment, priority/status matrix, staff E2E
8 Administration	user-management desktop/tablet/mobile captures	users-admin API, UserManagement UI, authorization and user-administration E2E; duplicate, invalid role, inactive, self, and last-admin guards
9 Zen Green/responsive	rendered UI spec plus terminal tokens/checklist/responsive/accessibility items 20–23	exact tokens, desktop >=992px, tablet 768–991px, mobile <768px, visible focus, labels/live errors, and no page-level overflow

Zen Green token record

Group	Approved values
Primary / secondary / pale green	#006B3C / #0B7A46 / #EAF6EF
Page / surface / primary text	#F5F7F6 / #FFFFFF / #1B2F24
Secondary text / border / read-only	#52665A / #D1DDD6 / #EEF3F0
Error / warning / urgent / low	#C53030 / #D69E2E / #E53E3E / #319795
Breakpoints	Desktop >=992px; tablet 768–991px; mobile <768px; no page-level horizontal overflow

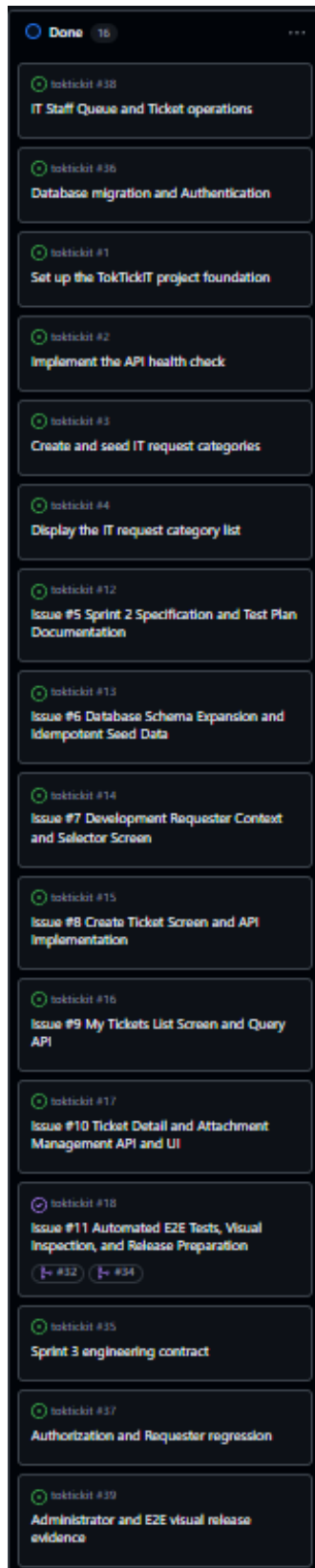
Evidence Appendix: local captures

The following pages retain the same Lab 2 evidence-page treatment while embedding the local source captures used for the final report. Items 1–18 document the workflow and source records; items 19–30 document the terminal verification; items 31–37 are final-checkout Playwright UI-state captures.

Range	Included evidence
1–18	Project board, main history, contract/release reviews, reviewer record, README, repository tree, specification/API/UI/test plans, and AI-use reflection
19–30	Final checkout, Zen Green tokens, visual checklist, responsive/accessibility rules, server/client tests, builds, Playwright installation/E2E, migration and seed
31–37	Invalid login, first-login gate, requester resolution, queue no-results, staff transition confirmation, administrator create form, duplicate-email error

All screenshot evidence below is retained from local runs or supplied final evidence captures. The report does not count screenshots as tests; test claims remain tied to executable paths and terminal results.

Evidence 01 — Project board with completed Lab 1–3 issues



Local verification capture retained in the final evidence set.

Evidence 02 — Main-branch commit history for the reviewed Lab 3 release

The screenshot displays the GitHub commit history for the repository `HolyThiccDaddy / laktikiti`. The interface is in dark mode. At the top, navigation links include Code, Issues (15), Pull requests, Agents, Discussions, Actions, Projects, Wiki, Security and quality, Insights, and Settings. A search bar is located in the top right corner.

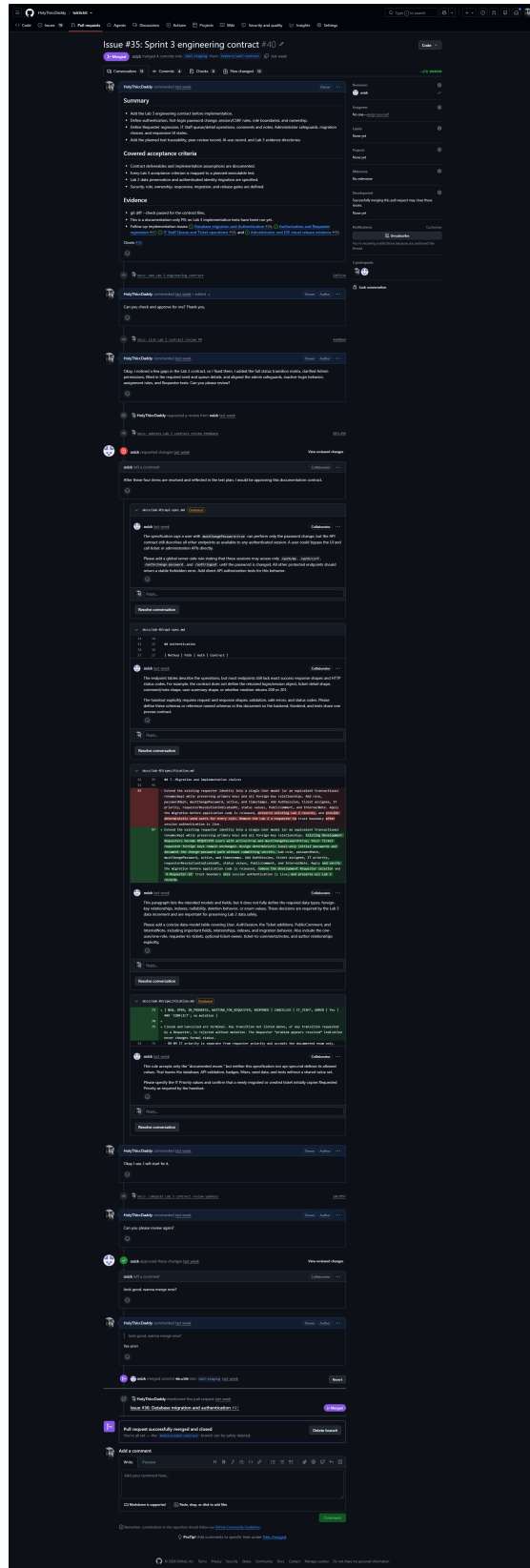
The main section is titled "Commits" and shows a list of commits grouped by date. The "main" branch is selected. The commits are as follows:

- Commits on Sep 19, 2025**
 - Merge pull request #47 from HolyThiccDaddy/feature/lab3-final-evidence (Verified, e798d3)
 - doc: match Lab 3 report to submission format (d6d433)
 - doc: finalize Lab 3 main release evidence (64d8d81)
 - Merge pull request #46 from HolyThiccDaddy/feature/lab3-final-release (Verified, 82ac36)
 - test: wait for ticket references before submitting (3b050a2)
 - Merge pull request #45 from HolyThiccDaddy/feature/39-admin-e2e-release (Verified, e82ac21)
 - doc: record PR 44 review and reflection (b699d08)
 - doc: align Lab 3 evidence records (7baf00)
 - doc: update Lab 3 review status (5c039d0)
- Commits on Sep 16, 2025**
 - Merge pull request #44 from HolyThiccDaddy/feature/39-admin-e2e-release (Verified, bcd572a)
 - test: cover administrator update races (Verified, 41562a8)
 - fix: prevent stale administrator user updates (Verified, 389f266)
- Commits on Sep 17, 2025**
 - fix: harden administrator management concurrency and filtering (b1dc978)
 - feat: complete Issue #39 admin and release evidence (78b6b04)
 - Merge pull request #43 from HolyThiccDaddy/feature/38-staff-queue (Verified, 4e07f08)
 - fix: address staff queue review feedback (799b6da)
 - feat: implement Issue #38 staff queue and ticket operations (92d057d)
- Commits on Sep 15, 2025**
 - Merge pull request #42 from HolyThiccDaddy/feature/37-requester-regression (Verified, 0e0002)
 - doc: refresh Lab 3 test totals (3b3d31a)
 - fix: address Lab 3 session and resolution review feedback (e9f9c20)
 - feat: migrate requester flows to authenticated sessions (09c0bc)
- Commits on Sep 12, 2025**
 - Merge pull request #41 from HolyThiccDaddy/feature/36-db-auth (Verified, b48db2)
 - doc: record Lab 3 implementation review (d0c236f)
 - fix: normalize first-login gate paths (9e0d73a)
 - fix: harden Lab 3 seed migration for legacy users (82c5aa)
 - fix: harden Lab 3 auth migration evidence (2d75d6)
 - feat: implement Lab 3 database migration and authentication (9c053f)
 - Merge pull request #40 from HolyThiccDaddy/feature/lab3-contract (Verified, 40ca398)
 - doc: complete Lab 3 contract review updates (24c7857)
 - doc: address Lab 3 contract review feedback (887c396)
 - doc: link Lab 3 contract review PR (4c98d0)
 - doc: add Lab 3 engineering contract (2a5510a)
- Commits on Sep 6, 2025**
 - Merge pull request #34 from HolyThiccDaddy/lab2-staging (Verified, d8e953)
 - Merge pull request #33 from HolyThiccDaddy/feature/lab2-reviewer-docs (Verified, 34da9d8)
 - doc: reconcile Lab 2 release evidence (95039b)

At the bottom, there are links for "Previous" and "Next". The footer contains copyright information: "© 2025 GitHub, Inc. Terms Privacy Security Status Community Docs Contact Manage cookies Do not share my personal information".

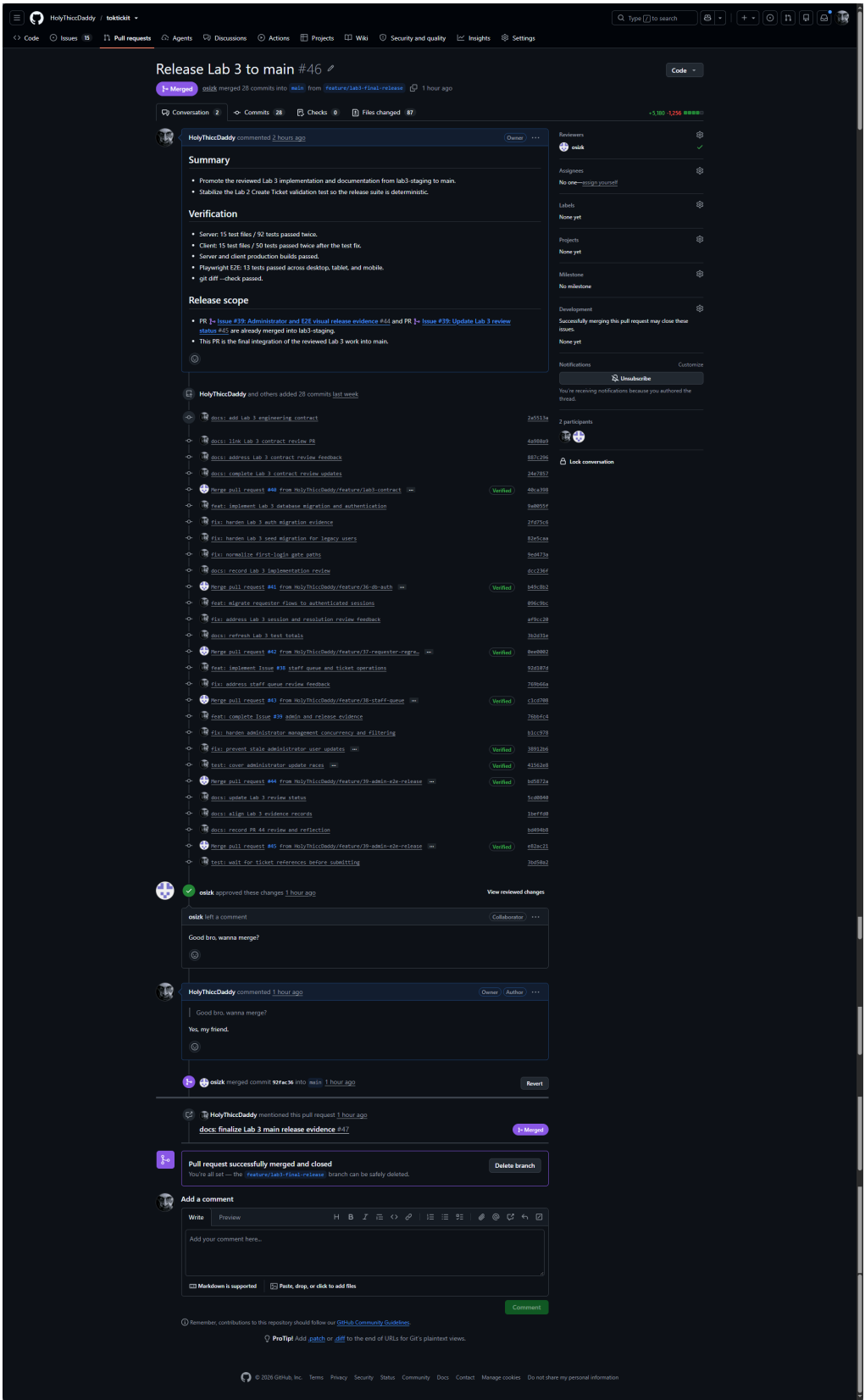
Local verification capture retained in the final evidence set.

Evidence 03 — PR #40 contract discussion, review feedback, and approval



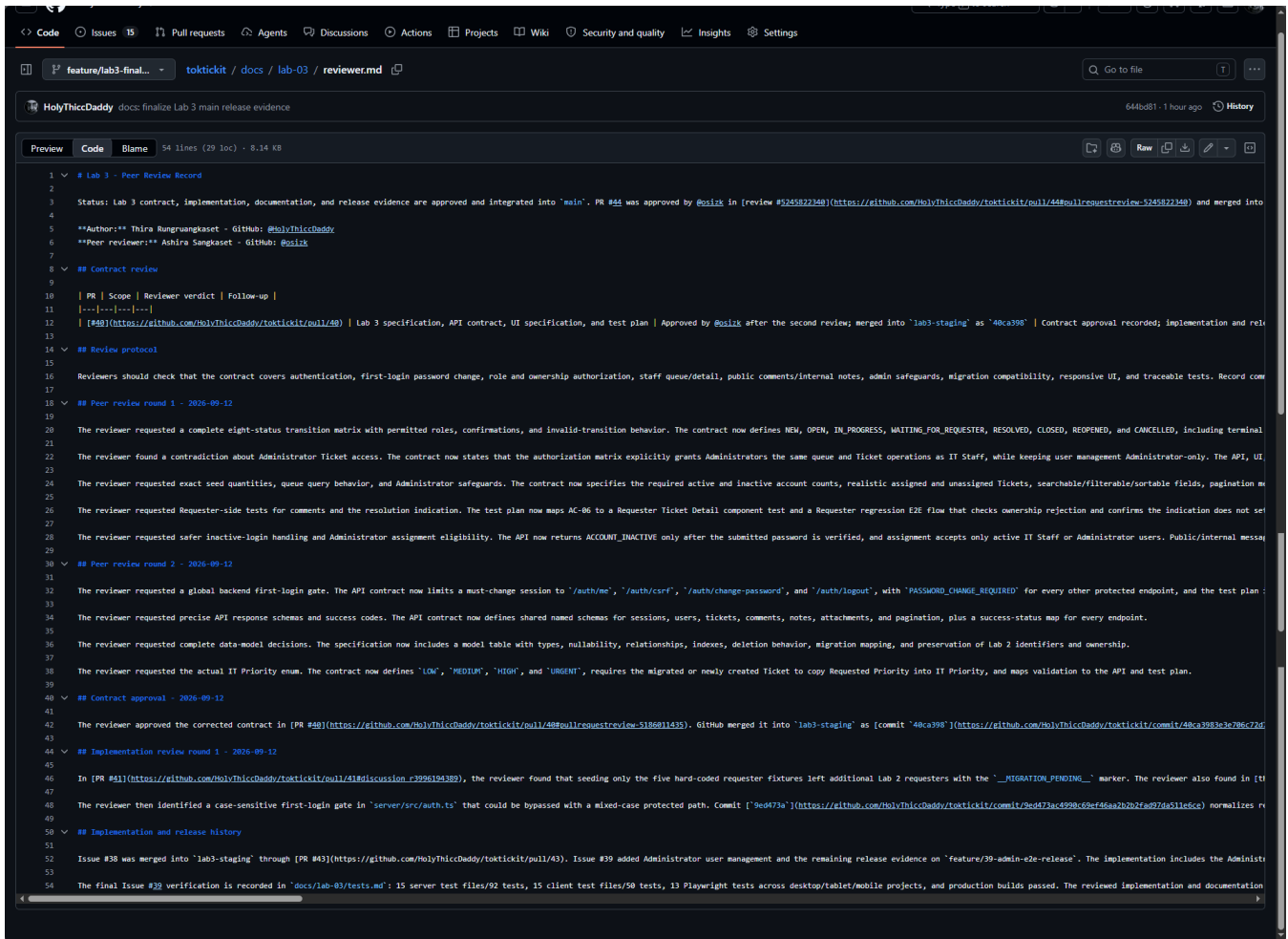
Local verification capture retained in the final evidence set.

Evidence 04 — PR #46 final release approval and merge into main



Local verification capture retained in the final evidence set.

Evidence 05 — Rendered peer-review record with review rounds and approvals



The screenshot displays a GitHub repository interface for the repository `feature/lab3-final...` under the path `toktickit / docs / lab-03 / reviewer.md`. The file is titled `docs: finalize Lab 3 main release evidence` and was last modified 1 hour ago. The file content is a detailed peer-review record for Lab 3, structured as follows:

- Lab 3 - Peer Review Record**
 - Status: Lab 3 contract, implementation, documentation, and release evidence are approved and integrated into 'main'. PR #44 was approved by @osik in [review #5245822348](https://github.com/HolyThiccDaddy/toktickit/pull/44#pullrequestreview-5245822348) and merged into
 - **Author:** Thira Rungruangkaset - GitHub: @HolyThiccDaddy
 - **Peer reviewer:** Ashira Sangkaset - GitHub: @osik
- Contract review**
 - PR | Scope | Reviewer verdict | Follow-up |
 - PR #40 [https://github.com/HolyThiccDaddy/toktickit/pull/40] | Lab 3 specification, API contract, UI specification, and test plan | Approved by @osik after the second review; merged into 'lab3-staging' as '40ca398' | Contract approval recorded; implementation and rel
- Review protocol**

Reviewers should check that the contract covers authentication, first-login password change, role and ownership authorization, staff queue/detail, public comments/internal notes, admin safeguards, migration compatibility, responsive UI, and traceable tests. Record com
- Peer review round 1 - 2026-09-12**

The reviewer requested a complete eight-status transition matrix with permitted roles, confirmations, and invalid-transition behavior. The contract now defines NEW, OPEN, IN_PROGRESS, WAITING_FOR_REQUESTER, RESOLVED, CLOSED, REOPENED, and CANCELLED, including terminal

The reviewer found a contradiction about Administrator Ticket access. The contract now states that the authorization matrix explicitly grants Administrators the same queue and Ticket operations as IT Staff, while keeping user management Administrator-only. The API, UI

The reviewer requested exact seed quantities, queue query behavior, and Administrator safeguards. The contract now specifies the required active and inactive account counts, realistic assigned and unassigned Tickets, searchable/filterable/sortable fields, pagination m

The reviewer requested Requester-side tests for comments and the resolution indication. The test plan now maps AC-86 to a Requester Ticket Detail component test and a Requester regression E2E flow that checks ownership rejection and confirms the indication does not set

The reviewer requested safer inactive-login handling and Administrator assignment eligibility. The API now returns ACCOUNT_INACTIVE only after the submitted password is verified, and assignment accepts only active IT Staff or Administrator users. Public/internal messa
- Peer review round 2 - 2026-09-12**

The reviewer requested a global backend first-login gate. The API contract now limits a must-change session to `/auth/me`, `/auth/csrf`, `/auth/change-password`, and `/auth/logout`, with `PASSWORD_CHANGE_REQUIRED` for every other protected endpoint, and the test plan

The reviewer requested precise API response schemas and success codes. The API contract now defines shared named schemas for sessions, users, tickets, comments, notes, attachments, and pagination, plus a success-status map for every endpoint.

The reviewer requested complete data-model decisions. The specification now includes a model table with types, nullability, relationships, indexes, deletion behavior, migration mapping, and preservation of Lab 2 identifiers and ownership.

The reviewer requested the actual IT Priority enum. The contract now defines `LOW`, `MEDIUM`, `HIGH`, and `URGENT`, requires the migrated or newly created Ticket to copy Requested Priority into IT Priority, and maps validation to the API and test plan.
- Contract approval - 2026-09-12**

The reviewer approved the corrected contract in [PR #40](https://github.com/HolyThiccDaddy/toktickit/pull/40#pullrequestreview-5186811435). GitHub merged it into 'lab3-staging' as [commit '40ca398'](https://github.com/HolyThiccDaddy/toktickit/commit/40ca3983e786c72d
- Implementation review round 1 - 2026-09-12**

In [PR #41](https://github.com/HolyThiccDaddy/toktickit/pull/41#discussion_r3996194389), the reviewer found that seeding only the five hard-coded requester fixtures left additional Lab 2 requesters with the `__MIGRATION_PENDING__` marker. The reviewer also found in [t

The reviewer then identified a case-sensitive first-login gate in `server/src/auth.ts` that could be bypassed with a mixed-case protected path. Commit ['9e0473a'](https://github.com/HolyThiccDaddy/toktickit/commit/9e0473ac499b09ef46ea7b2b2f2fa071d511ed6c) normalizes r
- Implementation and release history**

Issue #38 was merged into 'lab3-staging' through [PR #43](https://github.com/HolyThiccDaddy/toktickit/pull/43). Issue #39 added Administrator user management and the remaining release evidence on 'feature/39-admin-e2e-release'. The implementation includes the Administ

The final Issue #38 verification is recorded in 'docs/lab-03/tests.md': 15 server test files/92 tests, 15 client test files/50 tests, 13 Playwright tests across desktop/tablet/mobile projects, and production builds passed. The reviewed implementation and documentation

Local verification capture retained in the final evidence set.

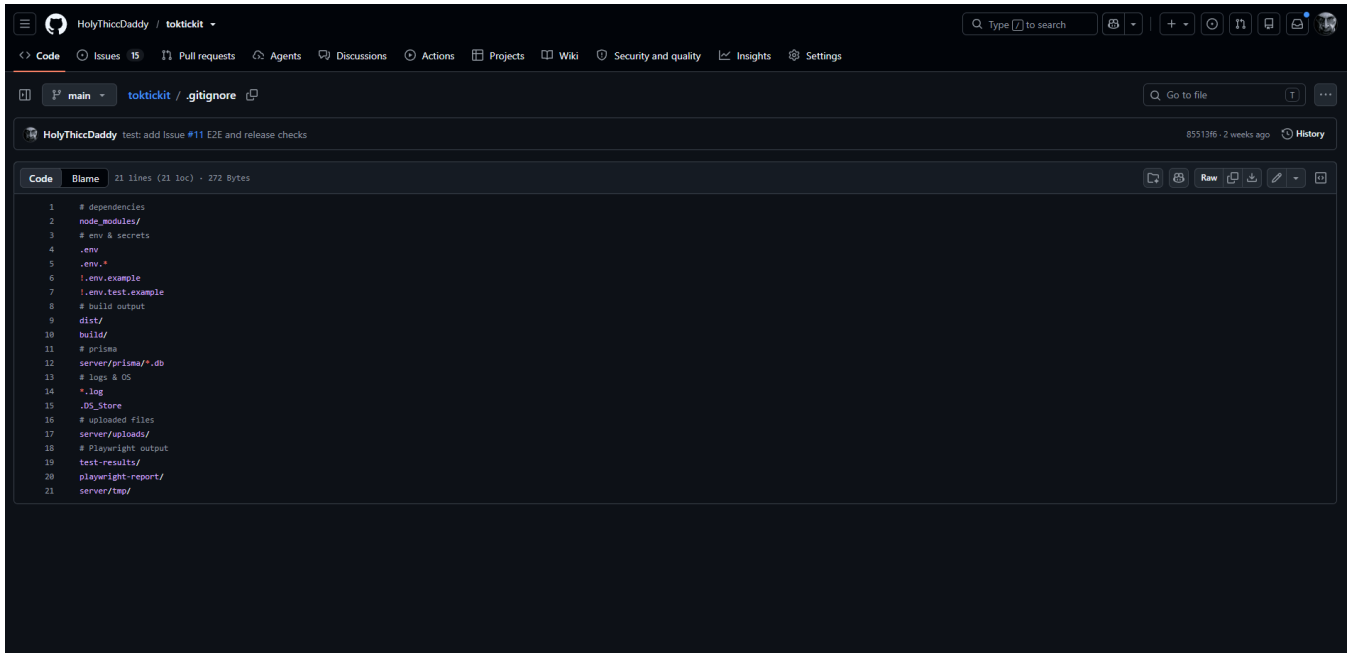
Evidence 06 — Rendered README with setup, verification, and document links

```

1  # taktickit - Lab 3 Authentication & Filtering
2
3  Taktickit is a full-stack IT service-desk application. Lab 3 replaces Lab 2's temporary Development Requester selector with authenticated sessions and adds the complete requester, IT Staff, and Administrator workflows. Requesters manage only their own tickets and attachments.
4
5  # Scope and Technology
6
7  **Frontend:** React 18, TypeScript, Vite, Bootstrap 5
8  **Backend:** Node.js, Express, TypeScript, Prisma ORM
9  **Database:** PostgreSQL
10 **Testing:** Vitest, Supertest, React Testing Library, Playwright
11 **Architecture:** Monorepo with 'client/', 'server/', and repository-root E2E specs
12
13 The released application includes the approved Lab 3 query, ticket detail, communication, and user-management workflow described in 'docs/lab-03/'.
14
15 # Repository structure
16
17 ```text
18 taktickit/
19 ├── client/                # React UI, component tests, and Playwright Abandonmy
20 ├── server/               # Express API, Prisma ORM/migrations, and API tests
21 ├── e2e/                  # Playwright journeys and deterministic setup/teardown
22 ├── docs/lab-02/          # Lab 2 contract, test plan, AI log, and review record
23 ├── docs/lab-03/          # Lab 3 contract, test plan, AI log, and review record
24 ├── artifacts/lab-02/     # Lab 2 test output and resumable/visual evidence
25 ├── artifacts/lab-03/     # Lab 3 resumable/visual evidence
26 ├── output/               # Final submission reports
27 ...
28
29 # Prerequisites
30
31 Node.js "20 or higher" (required by the current Playwright dependency)
32 PostgreSQL running locally on port "5432"
33 A PostgreSQL role that can create databases
34
35 # Installation and environment
36
37 ```bash
38 git clone https://github.com/HolyChicDaddy/taktickit.git
39 cd taktickit
40 cd server
41 npm install
42 copy .env.example .env
43 cd ../client
44 npm install
45
46 Set 'server.env' to a local development database, for example:
47
48 ```env
49 DATABASE_URL="postgresql://taktickit:taktickit@localhost:5432/taktickit-test-unique"
50 PORT=3000
51 TACTICKIT_API_URL_ROOT=/api/v1
52
53
54 Create the dedicated test database before running destructive suites:
55
56 ```bash
57 createdb -U taktickit taktickit_test
58 cd server
59 copy .env.test.example .env.test
60 npm run migrate:deploy
61 npm run prisma:seed
62 ...
63
64 'server.env.test' is local-only and ignored by Git. Test setup requires a database name ending in '_test' and writes attachments beneath a test-only temporary directory.
65
66 # Run the application
67
68 ```bash
69 # terminal 1
70 cd server
71 npm run dev
72
73 # terminal 2
74 cd client
75 npm run dev
76
77 The API is available at "http://localhost:3000" and Vite serves the UI at "http://localhost:5173".
78
79 # Verification commands
80
81 Run the smallest relevant suite first, then the full release checks:
82
83 ```bash
84 cd server
85 npm test
86 npm run build
87
88 cd ../client
89 npm test
90 npm run build
91 npm run playwright test
92
93 The Playwright command starts isolated services, resets deterministic fixtures in 'taktickit_test', checks requester, staff, and administrator journeys at desktop/tablet/mobile sizes, and writes JSOR results plus PMO evidence under the 'artifacts/' directories.
94
95 # API surface
96
97 - GET /api/health - service health
98 - POST /api/auth/login - create an authenticated session
99 - POST /api/auth/logout - revoke the current session
100 - GET /api/auth/me - return the current user summary
101 - GET /api/auth/reset - issue a CSRF token for state-changing requests
102 - POST /api/auth/change-password - simulate a first-time password change
103 - GET /api/categories - active ticket categories
104 - GET /api/active-systems - active related systems
105 - POST /api/tickets - create a 'new' ticket owned by the signed-in Requester
106 - GET /api/tickets - session-scoped search, filtering, sorting, and pagination
107 - GET /api/tickets/{id} - read-only ticket detail with public conversation
108 - POST /api/tickets/{id}/attachments - add a validated attachment to an owned ticket
109 - GET /api/attachments/{id}/download - download an active owned attachment
110 - DELETE /api/attachments/{id} - soft-remove an owned attachment with a reason
111 - GET/POST /api/tickets/{id}/comments - read or append public comments on an owned ticket
112 - POST /api/tickets/{id}/requester-resolve - record the Requester's resolvable indication
113 - GET/POST /api/staff/tickets/{id}/... - staff/queue assignment, IT priority, and status operations
114 - GET/POST /api/tickets/{id}/notes - internal notes for IT Staff and Administrators
115 - GET/POST /api/staff/assign-users... - Administrator user listing and safe account management
116
117 Protected identity is derived only from the server-side 'taktickit_session' cookie; client-supplied requester, owner, author, and role values are ignored. State-changing requests require the server-issued CSRF token. Attachment uploads enforce allowed type, extension,
118
119 # Lab 2 documentation
120
121 [Sprint specification](docs/lab-02/sprint-specification.md)
122 [NEXT API contract](docs/lab-02/api-contract.md)
123 [UI specification](docs/lab-02/ui-specification.md)
124 [Test plan and results](docs/lab-02/test-plan.md)
125 [AI use and reflection](docs/lab-02/ai-use.md)
126 [Peer-review record](docs/lab-02/peer-review.md)
127 [Lab 2 visual evidence](artifacts/lab-02/summary.md)
128
129 # Lab 3 documentation
130
131 [Sprint specification](docs/lab-03/sprint-specification.md)
132 [NEXT API contract](docs/lab-03/api-contract.md)
133 [UI specification](docs/lab-03/ui-specification.md)
134 [Test plan and results](docs/lab-03/test-plan.md)
135 [AI use and reflection](docs/lab-03/ai-use.md)
136 [Peer-review record](docs/lab-03/peer-review.md)
137 [Final submission report - user](docs/lab-03/ai-ai-final-submission-report-user.md)
138 [Final submission report - PMO](docs/lab-03/ai-ai-final-submission-report-pmo.md)
139
140 The core Lab 3 implementation was integrated into 'main' by release PR #22 ('a5f0b07'). The README/documentation update was prompted by PR #22 ('3d63f77'), and the final evidence/report update was prompted by PR #22 ('3d63f77').
141
142 Lab 3 was integrated into 'main' by Final Release PR #66 ('970c3c7') after peer-reviewed implementation PRs #22-#62 were merged into 'lab-staging'.
```

Local verification capture retained in the final evidence set.

Evidence 07 — Repository .gitignore evidence for secrets and generated files

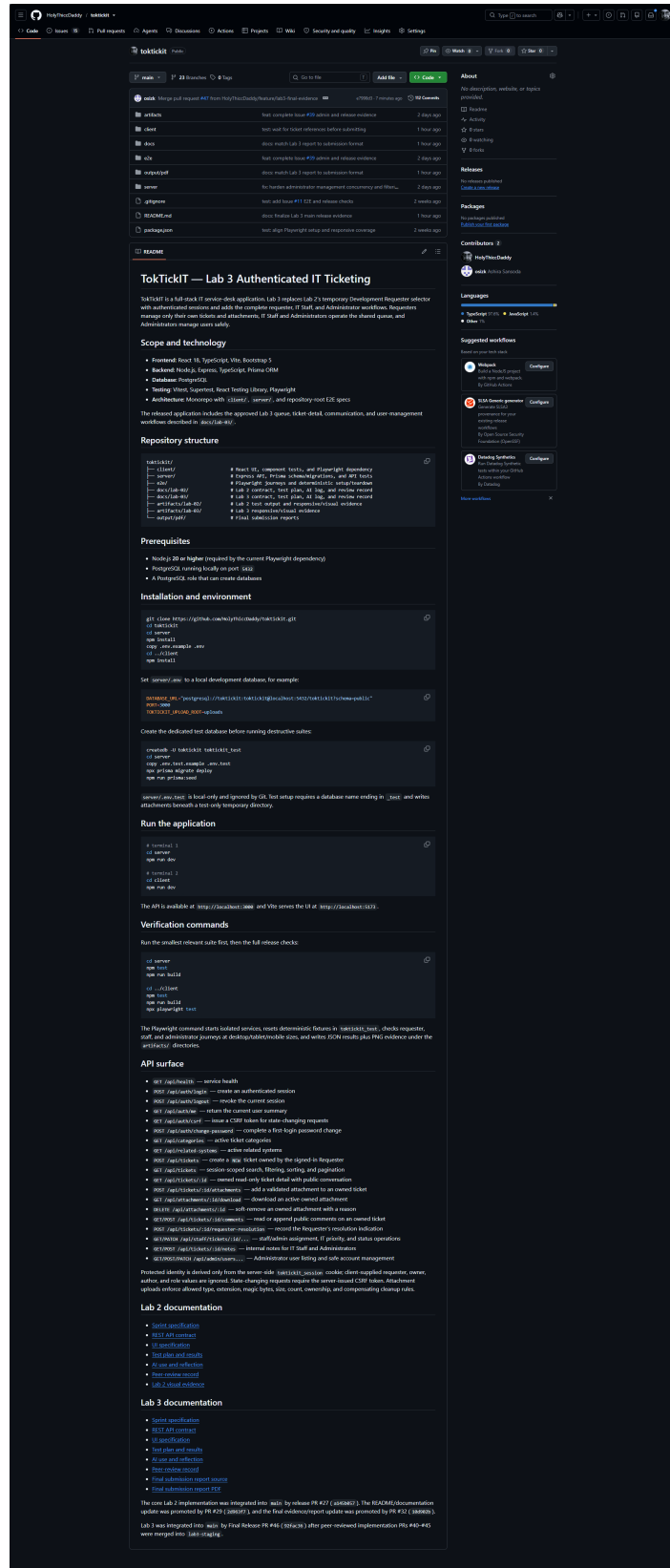


The screenshot shows a GitHub repository interface for 'HolyThiccDaddy / tokickit'. The file '.gitignore' is selected, showing its content. The file contains 21 lines of ignore rules. The repository has 15 issues, and the current view is for the 'main' branch. The file is 272 bytes and has 21 lines of code.

```
1 # dependencies
2 node_modules/
3 # env & secrets
4 .env
5 .env.*
6 !env.example
7 !env.test.example
8 # build output
9 dist/
10 build/
11 # prisma
12 server/prisma/*.db
13 # logs & OS
14 *.log
15 .DS_Store
16 # uploaded files
17 server/uploads/
18 # Playwright output
19 test-results/
20 playwright-report/
21 server/tmp/
```

Local verification capture retained in the final evidence set.

Evidence 08 — Main repository tree and README overview



Local verification capture retained in the final evidence set.

Evidence 09 — Rendered Lab 3 engineering specification

```
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999 1000
```

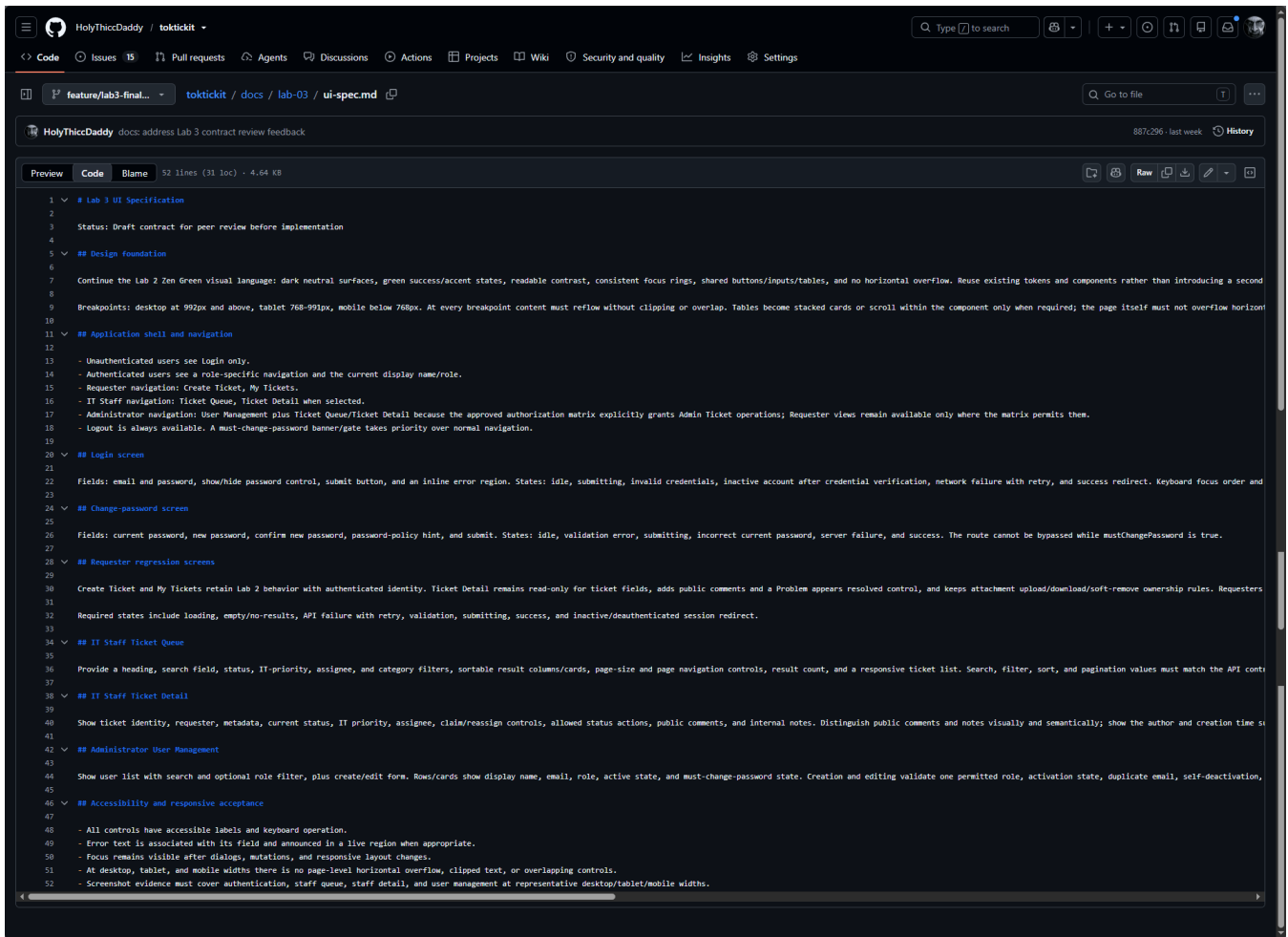
Local verification capture retained in the final evidence set.

Evidence 10 — Rendered API specification

```
1 # Lab 3 API Specification
2
3 Status: Approved contract (PR #88 merged into 'lab3-staging'); implementation is incremental by issue.
4
5 ## Conventions
6
7 - Base path: /api.
8 - JSON requests and responses use UTF-8 and the standard envelope '{ data: ... }' on success.
9 - Failures use '{ error: { code, message, fieldErrors? } }'. Codes are stable; messages never expose secrets, SQL, filesystem paths, storage keys, or unnecessary account details.
10 - Browser authentication uses an opaque HttpOnly cookie named 'toktickit_session'; the server resolves the user and role from the session.
11 - GET endpoints are safe. Every authenticated POST, PATCH, and DELETE requires the CSRF token returned by GET '/api/auth/csrf' in 'X-CSRF-Token'; the public login request is the exception.
12 - '401 UNAUTHENTICATED' means no valid session; '403 FORBIDDEN' means a valid session lacks the role or ownership; protected ticket resources may return '404 NOT_FOUND' for both missing and unauthorized records.
13 - Validation is '400 VALIDATION_ERROR'; duplicate or invalid state is '400 CONFLICT'; unexpected failures are '500 INTERNAL_ERROR'.
14 - A session with 'mustChangePassword=true' is authenticated but restricted. Until the password is changed, it may call only '/auth/me', '/auth/csrf', '/auth/change-password', and '/auth/logout'. Every other protected endpoint returns '403 PASSWORD_CHANGE_REQUIRED' with
15
16 ## Response schemas and success status codes
17
18 New Lab 3 JSON responses use the '{ data: ... }' envelope. The following named schemas are the shared contract for the backend, frontend, and tests. Legacy Lab 2 routes temporarily retain their existing response shapes during the incremental migration described below.
19
20 - 'UserSummary': '{ id: number, email: string, displayName: string, role: "REQUESTER" | "IT-STAFF" | "ADMIN", active: boolean, mustChangePassword: boolean }'.
21 - 'SessionData': '{ user: UserSummary, expiresAt: string }'; the server also sets or clears the HttpOnly session cookie as described above.
22 - 'PageMeta': '{ pages: number, pageSize: number, total: number, totalPages: number, sortBy: string, sortDir: "asc" | "desc" }'.
23 - 'CategorySummary': '{ id: number, name: string, description: string | null }'.
24 - 'RelatedSystemSummary': '{ id: number, name: string, description: string | null }'.
25 - 'AttachmentSummary': '{ id: number, originalFilename: string, mimeType: string, fileSize: number, isDeleted: boolean, createdAt: string }'; storage keys are never returned.
26 - 'TicketSummary': '{ id: number, ticketNumber: string, summary: string, requestedPriority: "LOW" | "MEDIUM" | "HIGH" | "URGENT", itPriority: "LOW" | "MEDIUM" | "HIGH" | "URGENT", currentStatus: "NEW" | "OPEN" | "IN_PROGRESS" | "WAITING_FOR_REQUESTER" | "RESOLVED" | "
27 - 'TicketDetail': 'TicketSummary' plus '{ description: string, category: CategorySummary, relatedSystem: RelatedSystemSummary, requesterResolutionIndicatedAt: string | null, attachments: AttachmentSummary[], publicComments: PublicComment[], internalNotes: InternalNote
28 - 'PublicComment': '{ id: number, ticketId: number, author: UserSummary, body: string, createdAt: string }'.
29 - 'InternalNote': '{ id: number, ticketId: number, author: UserSummary, body: string, createdAt: string }'.
30
31 Endpoint success contracts are:
32
33 | Method | Path | Success status and response |
34 |---|---|---|
35 | POST | '/auth/login' | '200 OK', '{ data: SessionData }'; a must-change user receives a restricted session |
36 | POST | '/auth/logout' | '204 No Content'; session cookie is cleared |
37 | GET | '/auth/me' | '200 OK', '{ data: UserSummary }' |
38 | GET | '/auth/csrf' | '200 OK', '{ data: { csrfToken: string, expiresAt: string } }' |
39 | POST | '/auth/change-password' | '200 OK', '{ data: UserSummary }' with 'mustChangePassword=false' |
40 | GET | '/categories' | '200 OK', '{ data: CategorySummary[] }' |
41 | GET | '/related-systems' | '200 OK', '{ data: RelatedSystemSummary[] }' |
```

Local verification capture retained in the final evidence set.

Evidence 11 — Rendered UI specification



The screenshot shows a code editor interface for a file named `ui-spec.md` located at `toktickit / docs / lab-03 /`. The editor displays a rendered UI specification document with the following content:

```
1 # Lab 3 UI Specification
2
3 Status: Draft contract for peer review before implementation
4
5 ## Design foundation
6
7 Continue the Lab 2 Zen Green visual language: dark neutral surfaces, green success/accnt states, readable contrast, consistent focus rings, shared buttons/inputs/tables, and no horizontal overflow. Reuse existing tokens and components rather than introducing a second
8
9 Breakpoints: desktop at 992px and above, tablet 768-991px, mobile below 768px. At every breakpoint content must reflow without clipping or overlap. Tables become stacked cards or scroll within the component only when required; the page itself must not overflow horizon
10
11 ## Application shell and navigation
12
13 - Unauthenticated users see Login only.
14 - Authenticated users see a role-specific navigation and the current display name/role.
15 - Requester navigation: Create Ticket, My Tickets.
16 - IT Staff navigation: Ticket Queue, Ticket Detail when selected.
17 - Administrator navigation: User Management plus Ticket Queue/Ticket Detail because the approved authorization matrix explicitly grants Admin Ticket operations; Requester views remain available only where the matrix permits them.
18 - Logout is always available. A must-change-password banner/gate takes priority over normal navigation.
19
20 ## Login screen
21
22 Fields: email and password, show/hide password control, submit button, and an inline error region. States: idle, submitting, invalid credentials, inactive account after credential verification, network failure with retry, and success redirect. Keyboard focus order and
23
24 ## Change-password screen
25
26 Fields: current password, new password, confirm new password, password-policy hint, and submit. States: idle, validation error, submitting, incorrect current password, server failure, and success. The route cannot be bypassed while mustChangePassword is true.
27
28 ## Requester regression screens
29
30 Create Ticket and My Tickets retain Lab 2 behavior with authenticated identity. Ticket Detail remains read-only for ticket fields, adds public comments and a Problem appears resolved control, and keeps attachment upload/download/soft-remove ownership rules. Requesters
31
32 Required states include loading, empty/no-results, API failure with retry, validation, submitting, success, and inactive/deauthenticated session redirect.
33
34 ## IT Staff Ticket Queue
35
36 Provide a heading, search field, status, IT-priority, assignee, and category filters, sortable result columns/cards, page-size and page navigation controls, result count, and a responsive ticket list. Search, filter, sort, and pagination values must match the API cont
37
38 ## IT Staff Ticket Detail
39
40 Show ticket identity, requester, metadata, current status, IT priority, assignee, claim/reassign controls, allowed status actions, public comments, and internal notes. Distinguish public comments and notes visually and semantically; show the author and creation time s
41
42 ## Administrator User Management
43
44 Show user list with search and optional role filter, plus create/edit form. Rows/cards show display name, email, role, active state, and must-change-password state. Creation and editing validate one permitted role, activation state, duplicate email, self-deactivation,
45
46 ## Accessibility and responsive acceptance
47
48 - All controls have accessible labels and keyboard operation.
49 - Error text is associated with its field and announced in a live region when appropriate.
50 - Focus remains visible after dialogs, mutations, and responsive layout changes.
51 - At desktop, tablet, and mobile widths there is no page-level horizontal overflow, clipped text, or overlapping controls.
52 - Screenshot evidence must cover authentication, staff Queue, staff detail, and user management at representative desktop/tablet/mobile widths.
```

Local verification capture retained in the final evidence set.

Evidence 12 — PR #40 changed-files view for contract artifacts

The screenshot displays the GitHub interface for Pull Request #40, titled "Sprint 3 engineering contract #40". The PR is merged and shows 4 commits. The "Files changed" tab is active, showing a list of files on the left and their diff on the right. The files include various .gitkeep files and documentation files like ai-use.md and api-spec.md. The diff for ai-use.md shows changes to the "Lab 3 AI Use and Reflection" section, and the diff for api-spec.md shows changes to the "Lab 3 API Specification" section.

Issue #35: Sprint 3 engineering contract #40

Merged [commit](#) merged 4 commits into [lab3-staging](#) from [Feature/lab3-contract](#) last week

Conversation 11 Commits 4 Checks 0 Files changed 13

0 / 13 viewed Submit comments

Filter files...

- artifacts/lab-03/screenshots
- authentication
- staff-queue
- staff-ticket-detail
- user-management
- client/tests/lab-03
- docs/lab-03
- ai-use.md
- api-spec.md
- reviewer.md
- specification.md
- tests.md
- ui-spec.md
- e2e/lab-03
- server/tests/lab-03
- e2e/lab-03
- server/tests/lab-03
- e2e/lab-03
- server/tests/lab-03

artifacts/lab-03/screenshots/authentication/.gitkeep

artifacts/lab-03/screenshots/staff-queue/.gitkeep

artifacts/lab-03/screenshots/staff-ticket-detail/.gitkeep

artifacts/lab-03/screenshots/user-management/.gitkeep

client/tests/lab-03/.gitkeep

docs/lab-03/ai-use.md

docs/lab-03/api-spec.md

```
1 + # Lab 3 AI Use and Reflection
16 + | 8 | Audit release evidence | Check source, logs, screenshots, and final report consistency | To be recorded before release PR |
17 +
18 + ## Reflection
19 +
20 + AI output is treated as a draft. Contract decisions remain subject to the Lab 3 handout and peer review. Security-sensitive behavior is accepted only when executable tests and real terminal evidence confirm it; suggestions that weaken server-side authorization, expose credentials, or invent unsupported requirements are rejected.
```

docs/lab-03/api-spec.md

```
1 + # Lab 3 API Specification
2 +
3 + Status: Draft contract for peer review before implementation
4 +
5 + ## Conventions
6 +
7 + - Base path: /api.
8 + - JSON requests and responses use UTF-8 and the standard envelope { data: ... } on success.
9 + - Failures use { error: { code, message, fieldErrors? } }. Codes are stable; messages never expose secrets, SQL, filesystem paths, storage keys, or unnecessary account details.
10 + - Browser authentication uses an opaque HttpOnly cookie named 'tokikit_session'; the server resolves the user and role from the session.
11 + - GET endpoints are safe. Every authenticated POST, PATCH, and DELETE requires the CSRF token returned by GET '/api/auth/csrf' in 'X-CSRF-Token'; the public login request is the exception.
12 + - '401 UNAUTHENTICATED' means no valid session; '403 FORBIDDEN' means a valid session lacks the role or ownership; protected ticket resources may return '404 NOT_FOUND' for both missing and unauthorized records.
13 + - Validation is '400 VALIDATION_ERROR'; duplicate or invalid state is '409 CONFLICT'; unexpected failures are '500 INTERNAL_ERROR'.
14 + - A session with 'mustChangePassword=true' is authenticated but restricted. Until the password is changed, it may call only '/auth/me', '/auth/csrf', '/auth/change-password', and '/auth/logout'. Every other protected endpoint returns '403 PASSWORD_CHANGE_REQUIRED' without performing a mutation.
```

Local verification capture retained in the final evidence set.

Evidence 13 — PR #40 commit history

The screenshot shows the GitHub web interface for a repository named 'HolyThiccDaddy / tolticket'. The top navigation bar includes links for Code, Issues (15), Pull requests (1), Agents, Discussions, Actions, Projects, Wiki, Security and quality, Insights, and Settings. A search bar is located on the right. The main content area displays 'Issue #35: Sprint 3 engineering contract #40' with a 'Code' button. Below the issue title, it states 'osix merged 4 commits into 'lab3-staging' from 'feature/lab3-contract' last week'. A tabbed interface shows 'Conversation' (11), 'Commits' (4), 'Checks' (0), and 'Files changed' (13). The 'Commits' tab is active, showing a list of four commits on Sep 12, 2026:

Commit Message	Commit Hash	Author
docs: add Lab 3 engineering contract	2a5513a	HolyThiccDaddy
docs: link Lab 3 contract review PR	4e988a9	HolyThiccDaddy
docs: address Lab 3 contract review feedback	887c296	HolyThiccDaddy
docs: complete Lab 3 contract review updates	24e7857	HolyThiccDaddy

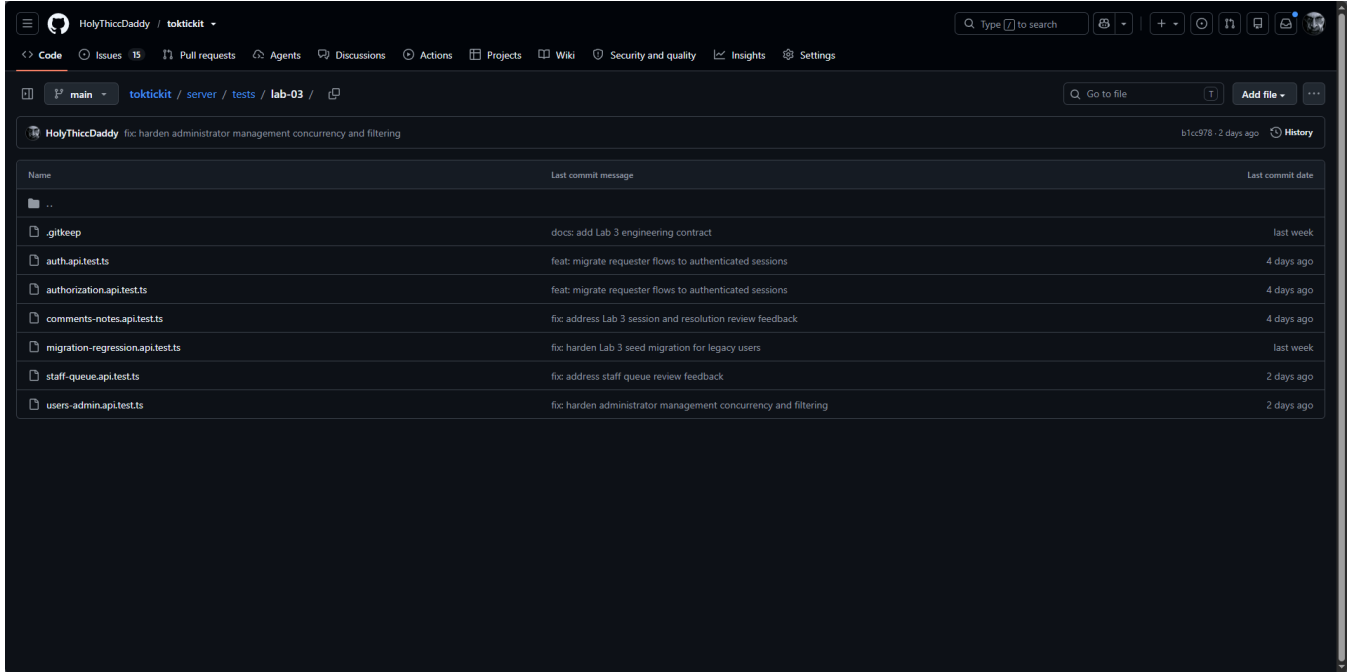
Local verification capture retained in the final evidence set.

Evidence 14 — Rendered tests plan and results

```
HolyThiccDaddy / toktickit +
Code Issues 15 Pull requests Agents Discussions Actions Projects Wiki Security and quality Insights Settings
feature/lab3-final... toktickit / docs / lab-03 / tests.md
HolyThiccDaddy docs: finalize Lab 3 main release evidence 644db81 - 1 hour ago History
Preview Code Blame 81 lines (65 loc) - 10.8 KB
1 # Lab 3 Test Plan and Results
2
3 Status: Final Issue #22 implementation and responsive release evidence were verified from 'main' after Final Release PR #26 merged as commit '92fac36'.
4
5 ## 1. Testing strategy
6
7 Use unit, API/integration, UI component, UI style, responsive, security/authorization, migration/regression, and Playwright E2E layers. Every acceptance criterion in specification.md is mapped below. Test fixtures use server/.env.test, a database name ending in _test,
8
9 ## 2. Traceability matrix
10
11 | ID | Acceptance criterion | Layer | Planned test file | evidence |
12 |---|---|---|---|
13 | T-01 | AC-01 valid login and role shell | API + UI + E2E | server/tests/lab-03/auth.api.test.ts; client/tests/lab-03/login.test.ts; e2e/lab-03/authentication.spec.ts |
14 | T-02 | AC-02 invalid/inactive login safety | API + UI + E2E | auth.api.test.ts; login.test.ts; authentication.spec.ts |
15 | T-03 | AC-03 first-login password gate and direct API blocking | API + UI + security + E2E | auth.api.test.ts; authorization.api.test.ts; client/tests/lab-03/ChangePassword.test.ts; authentication.spec.ts |
16 | T-04 | AC-04 logout/expiry protection | API + security + E2E | auth.api.test.ts; authorization.api.test.ts; authentication.spec.ts |
17 | T-05 | AC-05 requester Lab 2 regression/ownership | API + UI + migration + E2E | server/tests/lab-03/authorization.api.test.ts; server/tests/lab-02/create-ticket.api.test.ts, my-tickets.api.test.ts, ticket-detail.api.test.ts; e2e/lab-02/multi-requester-isolation.spec.ts |
18 | T-06 | AC-06 requester public comment/resolution indication and ownership | API + UI + security + E2E | server/tests/lab-03/comments-notes.api.test.ts; client/tests/lab-03/RequesterTicketDetail.test.ts; e2e/lab-03/requester-regression.spec.ts |
19 | T-07 | AC-07 shared queue search/filters/sort/pagination/responsive states | API + UI + style + responsive + E2E | server/tests/lab-03/staff-queue.api.test.ts; client/tests/lab-03/StaffTicketQueue.test.ts; e2e/lab-03/requester-regression.spec.ts |
20 | T-08 | AC-08 claim/reassign/IT Priority enum/complete status matrix authorization | API + UI + security + E2E | server/tests/lab-03/staff-queue.api.test.ts; client/tests/lab-03/StaffTicketDetail.test.ts; e2e/lab-03/staff-ticket-flow.spec.ts |
21 | T-09 | AC-09 public vs internal visibility and append-only validation | API + UI + security + E2E | server/tests/lab-03/staff-queue.api.test.ts; client/tests/lab-03/StaffTicketDetail.test.ts; e2e/lab-03/staff-ticket-flow.spec.ts |
22 | T-10 | AC-10 user management and activation state | API + UI + E2E | server/tests/lab-03/users-admin.api.test.ts; client/tests/lab-03/UserManagement.test.ts; e2e/lab-03/user-administration.spec.ts |
23 | T-11 | AC-11 self-deactivation/self-role-change, last-admin, and deactivated-user safety | API + security + UI | users-admin.api.test.ts; authorization.api.test.ts; UserManagement.test.ts |
24 | T-12 | AC-12 migration/data preservation/depotment seed | Migration + regression | server/tests/lab-03/migration-regression.api.test.ts; migration terminal evidence |
25 | T-13 | AC-13 forged identity/wrong-role rejection | API + security | server/tests/lab-03/authorization.api.test.ts |
26 | T-14 | AC-14 responsive no-overflow behavior | UI style + responsive + E2E | client/tests/lab-03/ResponsiveLayout.test.ts; all Lab 3 E2E projects; artifacts/lab-03/screenshots |
27 | T-15 | AC-15 release gates | Regression + build + E2E | server/client test and build logs; e2e results; release checklist |
28
29 T-03 must call a protected Ticket and Admin endpoint directly while "mustChangePassword=true" and assert "403 PASSWORD_CHANGE_REQUIRED", including an uppercase or mixed-case path variant; only "/auth/me", "/auth/csrf", "/auth/change-password", and "/auth/logout" remain
30
31 T-06 must exercise a Requester-owned Ticket Detail, successful Public Comment submission, successful Problem Appears Resolved indication, rejection for a different Requester, and an assertion that the indication does not change formal status to RESOLVED or CLOSED.
32
33 ## 3. Planned server suites
34
35 - server/tests/lab-03/auth.api.test.ts: valid/invalid login, inactive-account response after password verification, logout, session, CSRF, password policy, first-login gate, expiry, and response schema/status assertions.
36 - server/tests/lab-03/authorization.api.test.ts: session-derived identity, role boundaries, ownership, deactivated users, must-change-password gate, and safe errors.
37 - server/tests/lab-03/staff-queue.api.test.ts: queue search/filter/sort/pagination, invalid query behavior, claim reassign, reassignment to active IT Staff/Admin, IT Priority enum validation, status confirmation, staff detail, and response metadata.
38 - server/tests/lab-03/comments-notes.api.test.ts: Requester/Staff/Admin public comments, internal notes, visibility, append-only behavior, length/whitespace validation, and ownership.
39 - server/tests/lab-03/users-admin.api.test.ts: admin CRUD-like operations, role validation, activation on create/edit, initial password, self-deactivation/self-role-change, duplicate email, and last-admin guards.
40 - server/tests/lab-03/migration-regression.api.test.ts: preserved Lab 2 records, data-model relationships, Requested-to-IT Priority copy, and deterministic repeated seed.
41
42 ## 4. Planned client and E2E suites
43
44 - client/tests/lab-03/Login.test.tsx, ChangePassword.test.tsx, and RequesterTicketDetail.test.tsx.
45 - client/tests/lab-03/StaffTicketQueue.test.tsx, StaffTicketDetail.test.tsx, UserManagement.test.tsx, and ResponsiveLayout.test.tsx.
46 - e2e/lab-03/authentication.spec.ts, requester-regression.spec.ts, staff-ticket-flow.spec.ts, and user-administration.spec.ts.
47 - Screenshots are captured from real runs under artifacts/lab-03/screenshots/authentication, staff-queue, staff-ticket-detail, and user-management.
48
49 ## 5. Release checklist
50
51 - [x] All planned tests implemented and passing.
52 - [x] Server and client builds pass.
53 - [x] Playwright desktop, tablet, and mobile projects pass.
54 - [x] Migration and repeated seed pass twice against isolated_test database.
55 - [x] Security checks cover missing session, forged identity, wrong role, ownership, CSRF, inactive user, and Administrator safety guards.
56 - [x] First-login sessions cannot bypass the password gate through direct API calls.
57 - [x] API response schemas, status codes, data-model decisions, and IT Priority enum are covered by tests.
58 - [x] Terminal logs and visual evidence are captured and linked.
59 - [x] Results below are updated only with commands actually run.
60
61 ## 6. Executed results
62
63 - Issue #26 authentication API: 'server/tests/lab-03/auth.api.test.ts' - 10 tests passed, including generic inactive-account failures, case-insensitive first-login API-gate bypass checks, malformed JSON handling, expired sessions, and CSRF expiry.
64 - Issue #26 migration and deterministic-fixture regression: 'server/tests/lab-03/migration-regression.api.test.ts' - 7 tests passed, including canonical ownership links, repeated-seed equality, preservation of Lab 2 reference/counter/ticket/attachment rows, a legacy re
65 - Issue #22 requester authorization and communication regression: 'server/tests/lab-03/authorization.api.test.ts' plus 'server/tests/lab-03/comments-notes.api.test.ts' - 11 tests passed, covering session-derived identity, forged-header rejection, role/ownership boundar
66 - Issue #22 staff queue and ticket operations: 'server/tests/lab-03/staff-queue.api.test.ts' - 35 tests passed, covering shared queue queries, invalid-query safety, atomic claim, eligible reassignment, IT Priority, every valid BR-08 transition, required-confirmation fi
67 - Issue #22 staff UI components: 'client/tests/lab-03/StaffTicketQueue.test.tsx' and 'StaffTicketDetail.test.tsx' - 4 tests passed, covering queue states/filters/page size/result count, mobile card sorting controls, retryable validation failure, claim, priority/status
68 - Issue #22 staff E2E: 'client npm run test:e2e -- --e2e/lab-03/staff-ticket-flow.spec.ts' - 2 tests passed on the desktop project, covering requester-created work entering the shared staff queue, opening staff detail, and mobile card sorting controls at 375px.
69 - Issue #22 Administrator API: 'server/tests/lab-03/users-admin.api.test.ts' - 4 tests passed, covering safe listing and role boundaries, create/edit/activation/initial password reset, duplicate email, inactive role, inactive login, self-safety rules, and concurrent la
70 - Issue #22 Administrator UI: 'client/tests/lab-03/UserManagement.test.tsx' - 6 tests passed, covering list/search controls, filtered mutation refresh, stale-search response protection, partial edit payloads, filter changes during a pending mutation, create/edit/activ
71 - Issue #22 E2E: 'client npm run test:e2e -- --e2e/lab-03 --project=desktop' - 6 tests passed, covering authentication, requester communication, staff queue/detail, and Administrator user management.
72 - Issue #22 responsive E2E: 'client npm run test:e2e -- --e2e/lab-03/responsive.visual.spec.ts --project=tablet --project=mobile' - 2 tests passed; each project captured authentication, User Management, Staff Queue, and Staff Ticket Detail evidence.
73 - Full server regression: 'server npm test' - 15 test files, 92 tests passed.
74 - Full client regression: 'client npm vitest run --maxWorkers=1 --minWorkers=1' - 15 test files, 50 tests passed, including Administrator User Management, mobile staff queue sorting, and authenticated requester recovery. The serial run is the verified result.
75 - Authenticated requester E2E baseline: the retained evidence records 6 tests passed across desktop, tablet, and mobile projects, including public comment/resolution flow; a later full rerun retained six functional passes but its desktop visual capture hit an existing
76 - Server and client production builds passed after the Issue #22 implementation.
77 - 'npm prism validate' passed.
78 - 'npm prism migrate deploy' applied the two Lab 3 migrations to the isolated 'toktickit-test' database.
79 - Final integration: PR #26 merged the reviewed Lab 3 branch into 'main' as commit '92fac36'; the release-candidate verification above was run against the same merged source.
80
81 Issue #22 evidence is stored under 'artifacts/lab-03/screenshots/authentication,user-management,staff-queue,staff-ticket-detail' with desktop, tablet, and mobile captures from the real Playwright runs above.
```

Local verification capture retained in the final evidence set.

Evidence 15 — Main-branch server Lab 3 test tree



HolyThiccDaddy / toktickit

main toktickit / server / tests / lab-03

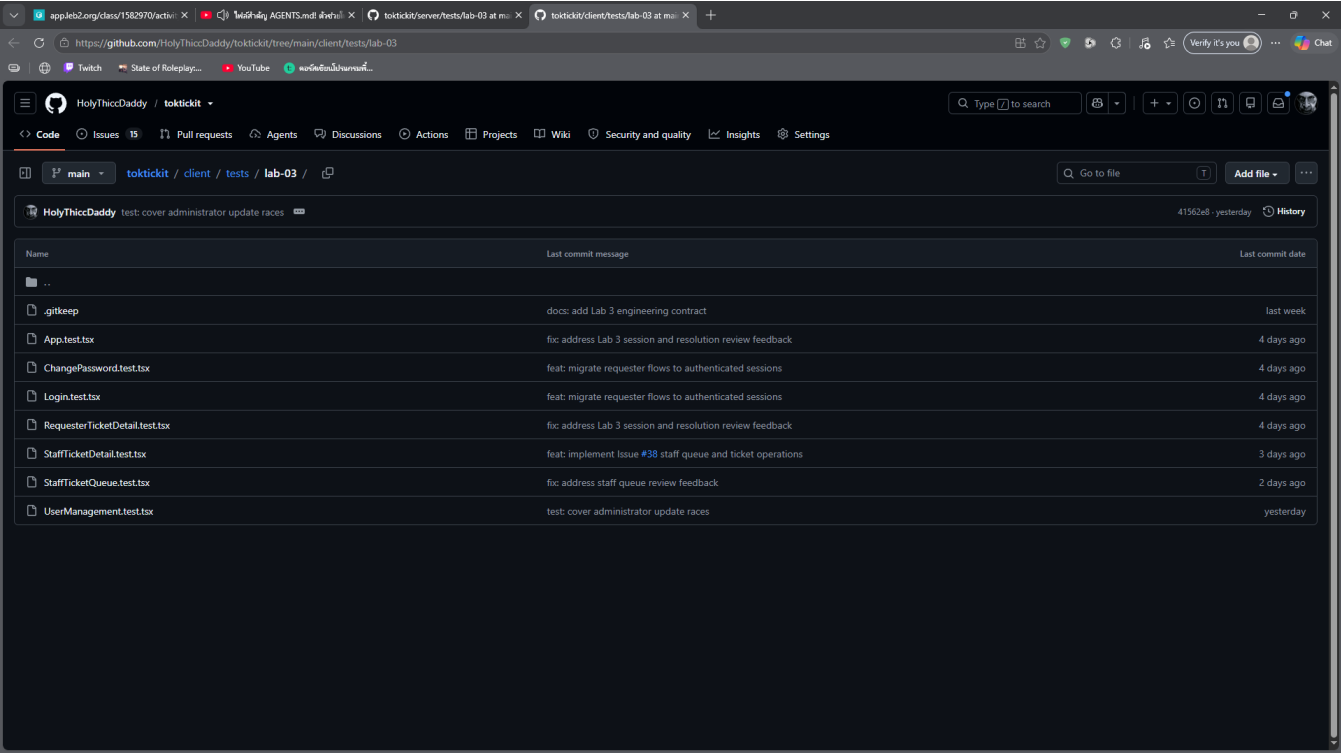
Go to file Add file

HolyThiccDaddy fix: harden administrator management concurrency and filtering b1cc978 · 2 days ago History

Name	Last commit message	Last commit date
..		
.gitkeep	docs: add Lab 3 engineering contract	last week
auth.api.test.ts	feat: migrate requester flows to authenticated sessions	4 days ago
authorization.api.test.ts	feat: migrate requester flows to authenticated sessions	4 days ago
comments-notes.api.test.ts	fix: address Lab 3 session and resolution review feedback	4 days ago
migration-regression.api.test.ts	fix: harden Lab 3 seed migration for legacy users	last week
staff-queue.api.test.ts	fix: address staff queue review feedback	2 days ago
users-admin.api.test.ts	fix: harden administrator management concurrency and filtering	2 days ago

Local verification capture retained in the final evidence set.

Evidence 16 — Main-branch client Lab 3 test tree



Local verification capture retained in the final evidence set.

Evidence 17 — Main-branch Playwright Lab 3 test tree

HolyThiccDaddy / toktickit

<> Code

Issues

Pull requests

Agents

Discussions

Actions

Projects

Wiki

Security and quality

Insights

Settings

main

toktickit

e2e

lab-03

Go to file

Add file

HolyThiccDaddy

feat: complete Issue #39 admin and release evidence

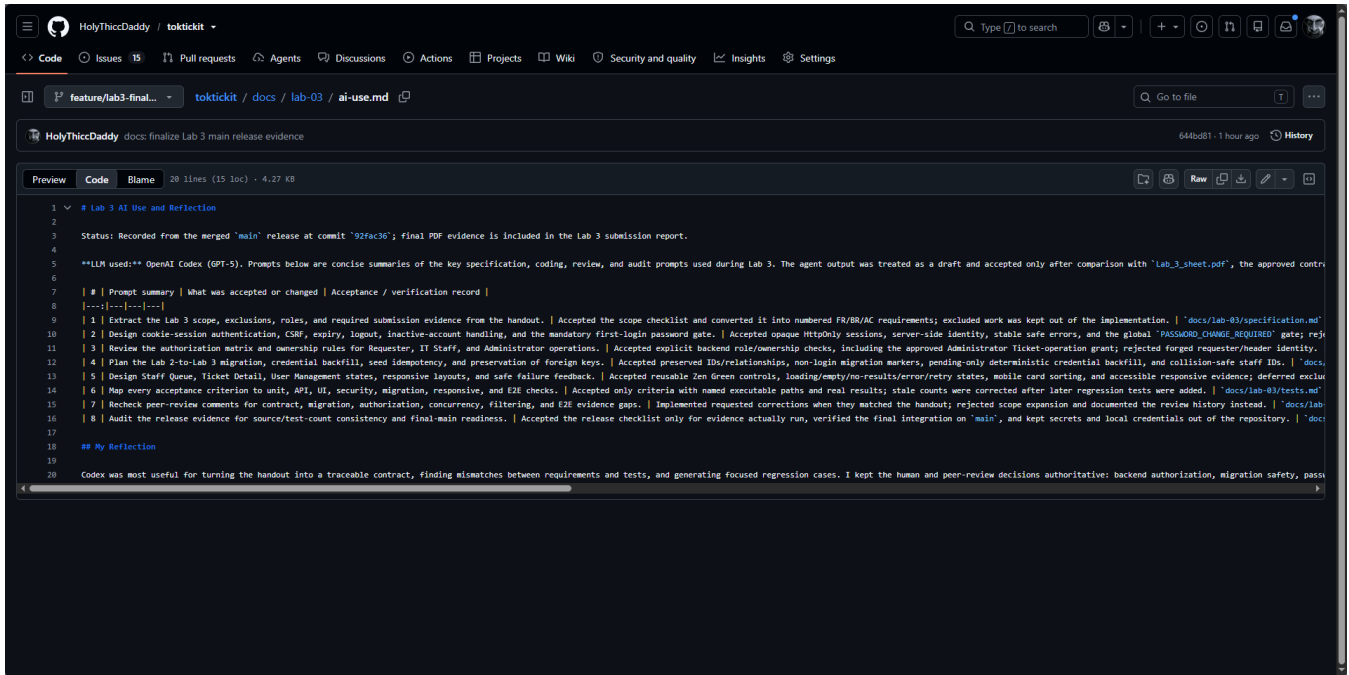
76bbf64 · 2 days ago

History

Name	Last commit message	Last commit date
..		
.gitkeep	docs: add Lab 3 engineering contract	last week
authentication.spects	feat: complete Issue #39 admin and release evidence	2 days ago
requester-regression.spects	feat: migrate requester flows to authenticated sessions	4 days ago
responsive.visual.spects	feat: complete Issue #39 admin and release evidence	2 days ago
staff-ticket-flow.spects	fix: address staff queue review feedback	2 days ago
user-administration.spects	feat: complete Issue #39 admin and release evidence	2 days ago

Local verification capture retained in the final evidence set.

Evidence 18 — Rendered AI-use record and reflection



The screenshot shows a code editor interface for a file named `ai-use.md` located at `docs/lab-03/`. The file content is as follows:

```
1 # Lab 3 AI Use and Reflection
2
3 Status: Recorded from the merged 'main' release at commit '92fac36'; final PDF evidence is included in the Lab 3 submission report.
4
5 **LLM used:** OpenAI Codex (GPT-5). Prompts below are concise summaries of the key specification, coding, review, and audit prompts used during Lab 3. The agent output was treated as a draft and accepted only after comparison with 'Lab_3_sheet.pdf', the approved contract.
6
7 | # | Prompt summary | What was accepted or changed | Acceptance / verification record |
8 |---|---|---|
9 | 1 | Extract the Lab 3 scope, exclusions, roles, and required submission evidence from the handout. | Accepted the scope checklist and converted it into numbered FR/BR/AC requirements; excluded work was kept out of the implementation. | 'docs/lab-03/specification.md'
10 | 2 | Design cookie-session authentication, CSRF, expiry, logout, inactive-account handling, and the mandatory first-login password gate. | Accepted opaque HttpOnly sessions, server-side identity, stable safe errors, and the global 'PASSWORD_CHANGE_REQUIRED' gate; rejected forged requester/header identity.
11 | 3 | Review the authorization matrix and ownership rules for Requester, IT Staff, and Administrator operations. | Accepted explicit backend role/ownership checks, including the approved Administrator Ticket-operation grant; rejected forged requester/header identity.
12 | 4 | Plan the Lab 2-to-Lab 3 migration, credential backfill, seed idempotency, and preservation of foreign keys. | Accepted preserved IDs/relationships, non-login migration markers, pending-only deterministic credential backfill, and collision-safe staff IDs. | 'docs/
13 | 5 | Design Staff Queue, Ticket Detail, User Management state, responsive layouts, and safe failure feedback. | Accepted reusable Zen Green controls, loading/empty/no-result/error/retry states, mobile card sorting, and accessible responsive evidence; deferred exclusion
14 | 6 | Map every acceptance criterion to unit, API, UI, security, migration, responsive, and E2E checks. | Accepted only criteria with named executable paths and real results; stale counts were corrected after later regression tests were added. | 'docs/lab-03/tests.md'
15 | 7 | Recheck peer-review comments for contract, migration, authorization, concurrency, filtering, and E2E evidence gaps. | Implemented requested corrections when they matched the handout; rejected scope expansion and documented the review history instead. | 'docs/lab
16 | 8 | Audit the release evidence for source/test-count consistency and final-main readiness. | Accepted the release checklist only for evidence actually run, verified the final integration on 'main', and kept secrets and local credentials out of the repository. | 'doc
17
18 ## My Reflection
19
20 Codex was most useful for turning the handout into a traceable contract, finding mismatches between requirements and tests, and generating focused regression cases. I kept the human and peer-review decisions authoritative: backend authorization, migration safety, pass
```

Local verification capture retained in the final evidence set.

Evidence 19 — Release-candidate checkout and branch/commit context

```
● PS D:\Downloads\toktickit> cd D:\Downloads\toktickit\.capture-main
● PS D:\Downloads\toktickit\.capture-main> git log -1 --oneline --decorate
e7998d3 (HEAD, origin/main) Merge pull request #47 from HolyThiccDaddy/feature/lab3-final-evidence
● PS D:\Downloads\toktickit\.capture-main> cd D:\Downloads\toktickit\.capture-main\server
PS D:\Downloads\toktickit\.capture-main\server> npm test

> toktickit-server@1.0.0 test
> vitest run

RUN v2.1.9 D:/Downloads/toktickit/.capture-main/server

✓ tests/lab-01/categories.test.ts (1)
✓ tests/lab-01/health.test.ts (1)
✓ tests/lab-02/attachment-validator.unit.test.ts (2)
✓ tests/lab-02/create-ticket.api.test.ts (4) 792ms
✓ tests/lab-02/my-tickets.api.test.ts (5) 652ms
✓ tests/lab-02/requesters.api.test.ts (1)
✓ tests/lab-02/seed.test.ts (6) 716ms
✓ tests/lab-02/ticket-detail.api.test.ts (4) 581ms
✓ tests/lab-02/ticket-number.unit.test.ts (1)
✓ tests/lab-03/auth.api.test.ts (10) 1467ms
✓ tests/lab-03/authorization.api.test.ts (5) 1209ms
✓ tests/lab-03/comments-notes.api.test.ts (6) 1142ms
✓ tests/lab-03/migration-regression.api.test.ts (7) 1887ms
✓ tests/lab-03/staff-queue.api.test.ts (35) 9224ms
✓ tests/lab-03/users-admin.api.test.ts (4) 783ms

Test Files 15 passed (15)
Tests 92 passed (92)
Start at 18:39:17
Duration 28.76s (transform 252ms, setup 0ms, collect 4.96s, tests 18.70s, environment 3ms, prepare 1.77s)

● PS D:\Downloads\toktickit\.capture-main\server> npm run build

> toktickit-server@1.0.0 build
> tsc

○ PS D:\Downloads\toktickit\.capture-main\server> |
```

Local verification capture retained in the final evidence set.

Evidence 20 — Zen Green color tokens from the UI specification

```
PS D:\Downloads\toktickit> cd D:\Downloads\toktickit\capture-main
PS D:\Downloads\toktickit\capture-main> Write-Host "=== Zen Green tokens ===" -ForegroundColor Cyan
=== Zen Green tokens ===
PS D:\Downloads\toktickit\capture-main> $lines = Get-Content .\docs\lab-02\ui-spec.md
PS D:\Downloads\toktickit\capture-main> $lines[0..24]
# Lab 2 Zen Green UI Specification

## 1. Overview & Design Philosophy
TokTickIT Lab 2 establishes the Zen Green Theme, a cohesive, professional, and accessible design system for enterprise IT service desks. All screens in Lab 2 and subsequent labs strictly reuse these component primitives, color tokens, and layout guidelines.

---

## 2. Color Tokens & Style Variables

| Token Name | Hex Value | Purpose / Usage |
| :--- | :--- | :--- |
| Primary Green | #00683C | Application header, primary action buttons, strong brand emphasis |
| Secondary Green | #007A46 | Active navigation tabs, focus rings, link hover states, subheaders |
| Pale Green | #EAF6EF | Selected rows, subtle section highlights, success container background |
| Page Background | #F5F7F6 | Default body/page canvas background |
| Surface / Card | #FFFFFF | Form cards, table containers, modal surfaces with subtle shadow |
| Text Primary | #1B2E24 | Dark charcoal-green for high-contrast, comfortable typography |
| Text Secondary | #52665A | Subtitles, helper text, table column headers, timestamps |
| Border Neutral | #D1DDD6 | Form input borders, table row dividers, card borders |
| Read-Only Shading | #EEF3F0 | Distinct background for non-editable input fields |
| Error Dark Red | #C53830 | Validation error text, field error borders, destructive actions |
| Error Background | #FFF5F5 | Inline error alert container background |
| Warning / Priority | #D69E2E | Medium/High priority badges, disclaimer banners |
| Urgent Priority | #E53E3E | Urgent priority badge |
| Low Priority | #319795 | Low priority badge |
```

Local verification capture retained in the final evidence set.

Evidence 21 — Visual checklist

```
PS D:\Downloads\toktickit> Write-Host "=== Visual checklist ===" -ForegroundColor Cyan
● === Visual checklist ===
PS D:\Downloads\toktickit> $lines = Get-Content .\docs\lab-02\tests.md
● PS D:\Downloads\toktickit> $lines[87..95]
● ## 4. Responsive and Visual Checklist

- [x] **Zen Green Color Palette:** Header uses `#006B3C`, hover/active accents use `#0B7A46`, pale background uses `#EAF6EF`, body uses `#F5F7F6`.
- [x] **Typography & Spacing:** Standardized font scale, consistent form input heights (38px), labels positioned above controls with red asterisk markers.
- [x] **Button Hierarchy:** Primary green buttons for main actions, outline buttons for secondary actions, amber for warnings, red for destructive soft-removal.
- [x] **Desktop Viewport (>= 992px):** Centered layout with max-width container, multi-column form, full data table with sortable headers.
- [x] **Tablet Viewport (768px - 991px):** Two-column form layout, responsive table layout with horizontal scrolling or condensed columns.
- [x] **Mobile Viewport (< 768px):** Single-column stacked form, ticket cards replacing table rows, touch-friendly buttons >= 44px height.
- [x] **No Visual Defects:** No clipped text, overlapping elements, or unexpected horizontal scrollbars at any breakpoint.
○ PS D:\Downloads\toktickit> |
```

Local verification capture retained in the final evidence set.

Evidence 22 — Responsive rules

```
PS D:\Downloads\toktickit> Write-Host "=== Responsive rules ===" -ForegroundColor Cyan
=== Responsive rules ===
• PS D:\Downloads\toktickit> $lines - Get-Content .\docs\lab-03\ui-spec.md
• PS D:\Downloads\toktickit> $lines[4..8]
• ## Design foundation
```

Continue the Lab 2 Zen Green visual language: dark neutral surfaces, green success/accent states, readable contrast, consistent focus rings, shared buttons/inputs/tables, and no horizontal overflow. Reuse existing tokens and components rather than introducing a second visual system.

Breakpoints: desktop at 992px and above, tablet 768px~991px, mobile below 768px. At every breakpoint content must reflow without clipping or overlap. Tables become stacked cards or scroll within the component only when required; the page itself must not overflow horizontally.

```
PS D:\Downloads\toktickit> █
```

Local verification capture retained in the final evidence set.

Evidence 23 — Accessibility and responsive acceptance

```
PS D:\Downloads\toktickit> Write-Host "=== Accessibility ===" -ForegroundColor Cyan
=== Accessibility ===
PS D:\Downloads\toktickit> $lines = Get-Content .\docs\lab-03\ui-spec.md
PS D:\Downloads\toktickit> $lines[45..51]
## Accessibility and responsive acceptance

- All controls have accessible labels and keyboard operation.
- Error text is associated with its field and announced in a live region when appropriate.
- Focus remains visible after dialogs, mutations, and responsive layout changes.
- At desktop, tablet, and mobile widths there is no page-level horizontal overflow, clipped text, or overlapping controls.
- Screenshot evidence must cover authentication, staff queue, staff detail, and user management at representative desktop/tablet/mobile widths.
PS D:\Downloads\toktickit> █
```

Local verification capture retained in the final evidence set.

Evidence 24 — Server npm test — 16 files / 92 tests passed after the suite split

```
PS D:\Downloads\toktickit> cd D:\Downloads\toktickit\worktree-main-audit\server
PS D:\Downloads\toktickit\worktree-main-audit\server> npm test

> toktickit-server@1.0.0 test
> vitest run

RUN v2.1.9 D:/Downloads/toktickit/worktree-main-audit/server

✓ tests/lab-01/categories.test.ts (1)
✓ tests/lab-01/health.test.ts (1)
✓ tests/lab-02/attachment-validator.unit.test.ts (2)
✓ tests/lab-02/create-ticket.api.test.ts (4) 676ms
✓ tests/lab-02/my-tickets.api.test.ts (5) 671ms
✓ tests/lab-02/requesters.api.test.ts (1)
✓ tests/lab-02/seed.test.ts (6) 488ms
✓ tests/lab-02/ticket-detail.api.test.ts (4) 548ms
✓ tests/lab-02/ticket-number.unit.test.ts (1)
✓ tests/lab-03/auth.api.test.ts (10) 1274ms
✓ tests/lab-03/authorization.api.test.ts (5) 986ms
✓ tests/lab-03/comments-notes.api.test.ts (6) 1128ms
✓ tests/lab-03/migration-regression.api.test.ts (7) 2032ms
✓ tests/lab-03/staff-queue.api.test.ts (4) 1484ms
✓ tests/lab-03/staff-ticket-detail.api.test.ts (31) 9043ms
✓ tests/lab-03/users-admin.api.test.ts (4) 871ms

Test Files 16 passed (16)
Tests      92 passed (92)
Start at   19:53:18
Duration   30.14s (transform 240ms, setup 0ms, collect 5.34s, tests 19.44s, environment 3ms, prepare 1.82s)
```

Local verification capture retained in the final evidence set.

Evidence 25 — Client npm test — 15 files / 50 tests passed

```
● PS D:\Downloads\toktickit\.capture-main\client> npm test

● > toktickit-client@1.0.0 test
  > vitest run
  ●
  RUN v2.1.9 D:/Downloads/toktickit/.capture-main/client
  ●
    ✓ tests/lab-01/App.test.tsx (3)
    ✓ tests/lab-02/api.test.tsx (7)
    ✓ tests/lab-02/CreateTicket.test.tsx (4) 512ms
    ✓ tests/lab-02/MyTickets.test.tsx (4) 463ms
    ✓ tests/lab-02/RequesterSelector.test.tsx (2)
    ✓ tests/lab-02/ResponsiveLayout.test.tsx (2)
    ✓ tests/lab-02/ThemeStyles.test.tsx (2)
    ✓ tests/lab-02/TicketDetail.test.tsx (6) 810ms
    ✓ tests/lab-03/App.test.tsx (2) 319ms
    ✓ tests/lab-03/ChangePassword.test.tsx (1)
    ✓ tests/lab-03/Login.test.tsx (3)
    ✓ tests/lab-03/RequesterTicketDetail.test.tsx (4) 353ms
    ✓ tests/lab-03/StaffTicketDetail.test.tsx (1) 625ms
    ✓ tests/lab-03/StaffTicketQueue.test.tsx (3) 713ms
    ✓ tests/lab-03/UserManagement.test.tsx (6) 1131ms

  Test Files 15 passed (15)
  Tests      50 passed (50)
  Start at   23:17:43
  Duration   7.22s (transform 590ms, setup 2.64s, collect 3.37s, tests 5.84s, environment 16.36s, prepare 2.67s)
```

Local verification capture retained in the final evidence set.

Evidence 26 — Server npm run build

```
● PS D:\Downloads\toktickit> cd D:\Downloads\toktickit\.capture-main
● PS D:\Downloads\toktickit\.capture-main> git log -1 --oneline --decorate
e7998d3 (HEAD, origin/main) Merge pull request #47 from HolyThiccDaddy/feature/lab3-final-evidence
● PS D:\Downloads\toktickit\.capture-main> cd D:\Downloads\toktickit\.capture-main\server
PS D:\Downloads\toktickit\.capture-main\server> npm test

> toktickit-server@1.0.0 test
> vitest run

RUN v2.1.9 D:/Downloads/toktickit/.capture-main/server

✓ tests/lab-01/categories.test.ts (1)
✓ tests/lab-01/health.test.ts (1)
✓ tests/lab-02/attachment-validator.unit.test.ts (2)
✓ tests/lab-02/create-ticket.api.test.ts (4) 792ms
✓ tests/lab-02/my-tickets.api.test.ts (5) 652ms
✓ tests/lab-02/requesters.api.test.ts (1)
✓ tests/lab-02/seed.test.ts (6) 716ms
✓ tests/lab-02/ticket-detail.api.test.ts (4) 581ms
✓ tests/lab-02/ticket-number.unit.test.ts (1)
✓ tests/lab-03/auth.api.test.ts (10) 1467ms
✓ tests/lab-03/authorization.api.test.ts (5) 1209ms
✓ tests/lab-03/comments-notes.api.test.ts (6) 1142ms
✓ tests/lab-03/migration-regression.api.test.ts (7) 1887ms
✓ tests/lab-03/staff-queue.api.test.ts (35) 9224ms
✓ tests/lab-03/users-admin.api.test.ts (4) 783ms

Test Files 15 passed (15)
Tests      92 passed (92)
Start at   18:39:17
Duration   28.76s (transform 252ms, setup 0ms, collect 4.96s, tests 18.70s, environment 3ms, prepare 1.77s)

● PS D:\Downloads\toktickit\.capture-main\server> npm run build

> toktickit-server@1.0.0 build
> tsc

○ PS D:\Downloads\toktickit\.capture-main\server> |
```

Local verification capture retained in the final evidence set.

Evidence 27 — Client npm run build

```
PS D:\Downloads\toktickit\capture-main\server> cd D:\Downloads\toktickit\capture-main\client
PS D:\Downloads\toktickit\capture-main\client> npm run build

> toktickit-client@1.0.0 build
> tsc && vite build

vite v6.4.3 building for production...
✓ 37 modules transformed.
dist/index.html          0.41 kB | gzip: 0.27 kB
dist/assets/index-B4cssW3.css 234.03 kB | gzip: 31.83 kB
dist/assets/index-DtMw3Mdc.js 220.95 kB | gzip: 61.26 kB
✓ built in 1.31s
PS D:\Downloads\toktickit\capture-main\client> npx playwright test

Running 13 tests using 1 worker

  ✓ 1 [desktop] › ..\e2e\lab-02\multi-requester-isolation.spec.ts:5:3 › Issue #37 multi-requester isolation › keeps requester A tickets invisible to requester B (2.0s)
  ✓ 2 [desktop] › ..\e2e\lab-02\requester-ticket-flow.spec.ts:8:3 › Issue #37 authenticated requester ticket flow › signs in, creates a ticket, opens detail, and removes an attachment (1.5s)
  ✓ 3 [desktop] › ..\e2e\lab-02\responsive.visual.spec.ts:30:1 › covers the requester journey across Create Ticket, My Tickets, and Ticket Detail (1.2s)
  ✓ 4 [desktop] › ..\e2e\lab-03\authentication.spec.ts:5:3 › Issue #39 authenticated release journey › completes first-login password setup and shows the Administrator shell (873ms)
  ✓ 5 [desktop] › ..\e2e\lab-03\requester-regression.spec.ts:5:3 › Issue #37 requester communication regression › adds a public comment and records Problem appears resolved without closing the ticket (2.1s)
  ✓ 6 [desktop] › ..\e2e\lab-03\responsive.visual.spec.ts:5:3 › Issue #39 responsive release evidence › captures authentication, user management, staff queue, and staff detail (2.3s)
  ✓ 7 [desktop] › ..\e2e\lab-03\staff-ticket-flow.spec.ts:5:3 › Issue #38 IT Staff queue and ticket operations › opens the shared queue and staff ticket detail with safe communication controls (1.5s)
  ✓ 8 [desktop] › ..\e2e\lab-03\staff-ticket-flow.spec.ts:31:3 › Issue #38 IT Staff queue and ticket operations › keeps queue sorting controls usable with mobile ticket cards (613ms)
  ✓ 9 [desktop] › ..\e2e\lab-03\user-administration.spec.ts:5:3 › Issue #39 Administrator user management › lists users and supports create, edit, activation, and initial-password reset (1.5s)
  ✓ 10 [tablet] › ..\e2e\lab-02\responsive.visual.spec.ts:30:1 › covers the requester journey across Create Ticket, My Tickets, and Ticket Detail (6.4s)
  ✓ 11 [tablet] › ..\e2e\lab-03\responsive.visual.spec.ts:5:3 › Issue #39 responsive release evidence › captures authentication, user management, staff queue, and staff detail (17.2s)
  ✓ 12 [mobile] › ..\e2e\lab-02\responsive.visual.spec.ts:30:1 › covers the requester journey across Create Ticket, My Tickets, and Ticket Detail (7.6s)
  ✓ 13 [mobile] › ..\e2e\lab-03\responsive.visual.spec.ts:5:3 › Issue #39 responsive release evidence › captures authentication, user management, staff queue, and staff detail (19.1s)

13 passed (1.2m)
PS D:\Downloads\toktickit\capture-main\client>
```

Local verification capture retained in the final evidence set.

```
PS D:\Downloads\toktickit> cd D:\Downloads\toktickit\client
● >> npx playwright install --list

Playwright version: 1.62.1
Browsers:
  C:\Users\KEN\AppData\Local\ms-playwright\chromium-1234
  C:\Users\KEN\AppData\Local\ms-playwright\chromium_headless_shell-1234
  C:\Users\KEN\AppData\Local\ms-playwright\ffmpeg-1011
  C:\Users\KEN\AppData\Local\ms-playwright\winldd-1007
References:
  D:\Downloads\toktickit\client\node_modules\playwright-core
  D:\Downloads\toktickit\node_modules\playwright-core
○ PS D:\Downloads\toktickit\client> █
```

Local verification capture retained in the final evidence set.

Evidence 29 — Playwright E2E — all 13 tests passed

```
PS D:\Downloads\toktickit\capture-main\server> cd D:\Downloads\toktickit\capture-main\client
PS D:\Downloads\toktickit\capture-main\client> npm run build

> toktickit-client@1.0.0 build
> tsc && vite build

vite v6.4.3 building for production...
✓ 37 modules transformed.
dist/index.html          0.41 kB | gzip: 0.27 kB
dist/assets/index-B4cssW3.css 234.03 kB | gzip: 31.83 kB
dist/assets/index-DtMw3Mdc.js 220.95 kB | gzip: 61.26 kB
✓ built in 1.31s
PS D:\Downloads\toktickit\capture-main\client> npx playwright test

Running 13 tests using 1 worker

  ✓ 1 [desktop] › ..\e2e\lab-02\multi-requester-isolation.spec.ts:5:3 › Issue #37 multi-requester isolation › keeps requester A tickets invisible to requester B (2.0s)
  ✓ 2 [desktop] › ..\e2e\lab-02\requester-ticket-flow.spec.ts:8:3 › Issue #37 authenticated requester ticket flow › signs in, creates a ticket, opens detail, and removes an attachment (1.5s)
  ✓ 3 [desktop] › ..\e2e\lab-02\responsive.visual.spec.ts:30:1 › covers the requester journey across Create Ticket, My Tickets, and Ticket Detail (1.2s)
  ✓ 4 [desktop] › ..\e2e\lab-03\authentication.spec.ts:5:3 › Issue #39 authenticated release journey › completes first-login password setup and shows the Administrator shell (873ms)
  ✓ 5 [desktop] › ..\e2e\lab-03\requester-regression.spec.ts:5:3 › Issue #37 requester communication regression › adds a public comment and records Problem appears resolved without closing the ticket (2.1s)
  ✓ 6 [desktop] › ..\e2e\lab-03\responsive.visual.spec.ts:5:3 › Issue #39 responsive release evidence › captures authentication, user management, staff queue, and staff detail (2.3s)
  ✓ 7 [desktop] › ..\e2e\lab-03\staff-ticket-flow.spec.ts:5:3 › Issue #38 IT Staff queue and ticket operations › opens the shared queue and staff ticket detail with safe communication controls (1.5s)
  ✓ 8 [desktop] › ..\e2e\lab-03\staff-ticket-flow.spec.ts:31:3 › Issue #38 IT Staff queue and ticket operations › keeps queue sorting controls usable with mobile ticket cards (613ms)
  ✓ 9 [desktop] › ..\e2e\lab-03\user-administration.spec.ts:5:3 › Issue #39 Administrator user management › lists users and supports create, edit, activation, and initial-password reset (1.5s)
  ✓ 10 [tablet] › ..\e2e\lab-02\responsive.visual.spec.ts:30:1 › covers the requester journey across Create Ticket, My Tickets, and Ticket Detail (6.4s)
  ✓ 11 [tablet] › ..\e2e\lab-03\responsive.visual.spec.ts:5:3 › Issue #39 responsive release evidence › captures authentication, user management, staff queue, and staff detail (17.2s)
  ✓ 12 [mobile] › ..\e2e\lab-02\responsive.visual.spec.ts:30:1 › covers the requester journey across Create Ticket, My Tickets, and Ticket Detail (7.6s)
  ✓ 13 [mobile] › ..\e2e\lab-03\responsive.visual.spec.ts:5:3 › Issue #39 responsive release evidence › captures authentication, user management, staff queue, and staff detail (19.1s)

13 passed (1.2m)
PS D:\Downloads\toktickit\capture-main\client>
```

Local verification capture retained in the final evidence set.

Evidence 30 — Test-database migration and two idempotent seed runs

```

PS D:\Downloads\toktickit> cd D:\Downloads\toktickit\capture-main\server
● PS D:\Downloads\toktickit\capture-main\server>
PS D:\Downloads\toktickit\capture-main\server> $line = Get-Content .env.test |
>> Where-Object { $_.StartsWith('DATABASE_URL=') } |
>> Select-Object -First 1
● PS D:\Downloads\toktickit\capture-main\server>
○ PS D:\Downloads\toktickit\capture-main\server> $testUrl = $line.Substring(13).Trim().Trim('')
● PS D:\Downloads\toktickit\capture-main\server>
● PS D:\Downloads\toktickit\capture-main\server> if (-not $testUrl.StartsWith('postgresql://') -and
>> -not $testUrl.StartsWith('postgres://')) {
>> throw "DATABASE_URL is not a PostgreSQL URL"
>> }
○ PS D:\Downloads\toktickit\capture-main\server>
● PS D:\Downloads\toktickit\capture-main\server> if (-not $testUrl.Contains('_test')) {
>> throw "DATABASE_URL is not a test database"
>> }
● PS D:\Downloads\toktickit\capture-main\server>
PS D:\Downloads\toktickit\capture-main\server> $env:DATABASE_URL = $testUrl
PS D:\Downloads\toktickit\capture-main\server> Write-Host "Test database loaded; URL hidden." -ForegroundColor Green
● Test database loaded; URL hidden.
○ PS D:\Downloads\toktickit\capture-main\server>
PS D:\Downloads\toktickit\capture-main\server> npx prisma migrate deploy
● Prisma schema loaded from prisma/schema.prisma
Datasource "db": PostgreSQL database "toktickit_test", schema "public" at "localhost:5432"

5 migrations found in prisma/migrations

No pending migrations to apply.
PS D:\Downloads\toktickit\capture-main\server> npm run prisma:seed
●
> toktickit-server@1.0.0 prisma:seed
> tsx prisma/seed.ts

Seeded 4 categories.
Seeded 7 related systems.
Seeded 5 development requesters (4 active, 1 inactive).
Initialized ticket counter.
● PS D:\Downloads\toktickit\capture-main\server> npm run prisma:seed

> toktickit-server@1.0.0 prisma:seed
> tsx prisma/seed.ts

Seeded 4 categories.
Seeded 7 related systems.
Seeded 5 development requesters (4 active, 1 inactive).
Initialized ticket counter.

```

Local verification capture retained in the final evidence set.

Evidence 31–37 — Invalid login with safe error and retry

TokTickIT

Sign in to IT Service Desk

Use your TokTickIT account to view and manage your requests.

Invalid email or password

Email

not-a-user@example.com

Password

.....

Show

[Retry sign in](#)

Sign in

Final-checkout Playwright state capture from the merged Lab 3 release.

Evidence 31–37 — First-login password-change gate

TokTickIT

Change your password

Your account requires a password change before you can access tickets.

Current password

New password

Use 12-128 characters.

Confirm new password

Update passwordSign out

Signed in as jennifer.anderson@example.com

Final-checkout Playwright state capture from the merged Lab 3 release.

Evidence 31–37 — Requester comment and resolution indication

TokTickIT

HomeMy Tickets+ Create Ticket

Jennifer Anderson
REQUESTERSign out

[← Back to My Tickets](#)

NEW

Ticket No.
TKT-2026-000001

TICKET DATE
9/27/2026, 6:13:59 PM

CATEGORY
Software

REQUESTED PRIORITY
MEDIUM

REQUESTER
Jennifer Anderson (jennifer.anderson@example.com)

RELATED SYSTEM
LEB2 App

IT PRIORITY
MEDIUM

Evidence state ticket

A real ticket used for final evidence state capture.

Attachments

0 of 5 active attachments

No active attachments.

Add Attachment

Public conversation

Marked as appears resolved

Indicated 9/27/2026, 6:14:00 PM

Jennifer Anderson · 9/27/2026, 6:14:00 PM

Evidence comment from the requester.

Add a public comment

Post comment

Final-checkout Playwright state capture from the merged Lab 3 release.

Evidence 31–37 — Staff queue filtered no-results state

TokTickIT

Ticket Queue

Alex Staff
IT_STAFF

Sign out

Ticket Queue

Shared queue for IT Staff.

IT_STAFF

Search queue

no-result-evidence-999

Status

All statuses

▼

IT Priority

All priorities

▼

Assignee ID

Any assignee

Category

All categories

▼

Sort by

Updated date

▼

Sort direction

Descending

▼

Results per page

20

▼

Clear

No matching tickets found.

Clear filters

Final-checkout Playwright state capture from the merged Lab 3 release.

Evidence 31-37 — Staff transition confirmation

TokTickIT

Ticket QueueTicket Detail

Alex Staff
IT_STAFFSign out

[← Back to queue](#)

NEW

Ticket No.

TKT-2026-000001

MEDIUM

REQUESTER
Jennifer Anderson (jennifer.anderson@example.com)

RELATED SYSTEM
LEB2 App

UPDATED
9/27/2026, 6:14:00 PM

CATEGORY
Software

CREATED
9/27/2026, 6:13:59 PM

Evidence state ticket

A real ticket used for final evidence state capture.

Ticket operations

Assignee ID

Blank to unassign

Claim for me

Save assignment

IT Priority

MEDIUM

Save

Move status

CANCELLED (confirmation required)

Confirm transition

Update status

Attachments

No attachments on this ticket.

Public comments

Jennifer Anderson · 9/27/2026, 6:14:00 PM
Evidence comment from the requester.

Add a public comment

Post public comment

Internal notes

Staff only

No internal notes yet.

Add an internal note

Add internal note

Final-checkout Playwright state capture from the merged Lab 3 release.

Evidence 31-37 — Administrator create-user form

TokTickIT

User ManagementTicket Queue

TokTickIT AdministratorADMINSign out

User Management

Manage roles and account activation without exposing credentials.

ADMIN

Search users

Role

All roles

Create user

Create user

Cancel

Email

Display name

Role

REQUESTER

☒ Active account

Initial password

12-128 characters; the user must change it on first login.

Create user

Final-checkout Playwright state capture from the merged Lab 3 release.

Evidence 31–37 — Duplicate-email safe error

TokTickIT

User Management

Ticket Queue

TokTickIT Administrator

ADMIN

Sign out

User Management

Manage roles and account activation without exposing credentials.

ADMIN

A user with this email already exists

Retry

Search users

Name or email

Role

All roles

Create user

Create user

[Cancel](#)

Email

admin@example.com

Display name

Duplicate evidence user

Role

REQUESTER

☒ Active account

Initial password

.....

12-128 characters; the user must change it on first login.

Create user

Display name	Email	Role	State	Actions
Alex Staff	alex.staff@example.com	IT_STAFF	Active	Edit Deactivate Reset password
Casey Staff	casey.staff@example.com	IT_STAFF	Active Password change required	Edit Deactivate Reset password
David Lee	david.lee@example.com	REQUESTER	Active Password change required	Edit Deactivate Reset password
Inactive Staff	inactive.staff@example.com	IT_STAFF	Inactive Password change required	Edit Activate Reset password
Jamie Staff	jamie.staff@example.com	IT_STAFF	Active Password change required	Edit Deactivate Reset password
Jennifer Anderson	jennifer.anderson@example.com	REQUESTER	Active	Edit Deactivate Reset password
Michael Brown	michael.brown@example.com	REQUESTER	Active Password change required	Edit Deactivate Reset password
Robert Taylor	robert.taylor@example.com	REQUESTER	Inactive Password change required	Edit Activate Reset password
Sarah Johnson	sarah.johnson@example.com	REQUESTER	Active Password change required	Edit Deactivate Reset password
TokTickIT Administrator	admin@example.com	ADMIN	Active	Edit Deactivate Reset password

Final-checkout Playwright state capture from the merged Lab 3 release.

Submission close-out

The primary deliverable is output/pdf/LAB_03_FINAL_SUBMISSION_REPORT.pdf. The identical _10-10 filename is retained as an alias. The core ten pages preserve the Lab 2 report template; this addendum adds the explicit master-spec crosswalk, final-main provenance, local verification, and final UI-state evidence.

The Staff Ticket Detail API suite is now an explicit server/tests/lab-03/staff-ticket-detail.api.test.ts file, matching the Lab 3 minimum repository structure. No unsupported requirement, credential, or hidden UI state is presented as evidence.